



# **Construction of a real time multivariate GARCH risk management tool for intra-day portfolio management**

**implemented via the Dash framework by Plotly in Python**

## **Master's Thesis**

**to acquire the academic title**

**“Master of Science (M.Sc.)”**

**in the degree International Finance**

**of the**

**Berlin School of Economics and Law**

**Supervisors:**      **Prof. Dr. Natalie Packham**

**Prof. Dr. Ursula Walther**

**Submission:**      **Berlin, 16.02.2022**

**Author:**            **Toni Meiners**

**Student ID No: 521026**

**s\_meinerst@stud.hwr-berlin.de**

**Word count:**      **15008**

## Table of Contents

|                                            |    |
|--------------------------------------------|----|
| Table of Contents .....                    | II |
| List of Abbreviations.....                 | IV |
| List of Figures.....                       | V  |
| List of Tables.....                        | VI |
| 1. Introduction .....                      | 1  |
| 2. Scope of research.....                  | 3  |
| 3. Theoretical background .....            | 4  |
| 3.1. GARCH models.....                     | 4  |
| 3.1.1. ARCH process .....                  | 4  |
| 3.1.2. GARCH process .....                 | 5  |
| 3.1.3. Multivariate GARCH .....            | 6  |
| 3.1.4. DCC multivariate GARCH.....         | 8  |
| 3.2. Intraday Seasonality .....            | 10 |
| 3.3. Model Evaluation .....                | 12 |
| 3.3.1. Univariate fit.....                 | 12 |
| 3.3.2. Multivariate fit.....               | 13 |
| 3.4. Value At Risk .....                   | 15 |
| 3.5. Historical Scenario .....             | 18 |
| 4. Concept and Design.....                 | 19 |
| 4.1. Software Development Life Cycle ..... | 19 |
| 4.2. Requirement Analysis .....            | 20 |
| 4.3. Design.....                           | 23 |
| 4.4. Code.....                             | 24 |
| 5. Implementation.....                     | 24 |
| 5.1. Sample .....                          | 25 |
| 5.1.1. Prices .....                        | 25 |

|        |                                |    |
|--------|--------------------------------|----|
| 5.1.2. | Volume .....                   | 26 |
| 5.2.   | Intraday Seasonality .....     | 26 |
| 5.3.   | DCC multivariate GARCH.....    | 28 |
| 5.4.   | Model Evaluation .....         | 32 |
| 5.5.   | Value at Risk .....            | 33 |
| 5.6.   | Historical Scenario .....      | 36 |
| 5.7.   | User Interface .....           | 36 |
| 6.     | Evaluation.....                | 38 |
| 6.1.   | Risk.....                      | 38 |
| 6.2.   | Performance.....               | 39 |
| 6.3.   | Dash.....                      | 40 |
| 7.     | Limitations.....               | 41 |
| 7.1.   | Modelling .....                | 41 |
| 7.1.1. | GARCH .....                    | 41 |
| 7.1.2. | Dimensionality .....           | 42 |
| 7.1.3. | Model Evaluation .....         | 43 |
| 7.1.4. | DCC multivariate GARCH.....    | 44 |
| 7.1.5. | Seasonality.....               | 44 |
| 7.1.6. | Value At Risk .....            | 45 |
| 7.2.   | Dash.....                      | 46 |
| 8.     | Further research .....         | 47 |
| 9.     | Conclusion.....                | 48 |
|        | Publication bibliography ..... | 49 |
|        | Appendix A .....               | 56 |
|        | Appendix B.....                | 59 |

## **List of Abbreviations**

|            |                                                             |
|------------|-------------------------------------------------------------|
| ACF        | Autocorrelation Function                                    |
| ARCH       | Auto Regressive Conditional Heteroskedastic                 |
| CDF        | Cumulative Distribution Function                            |
| DCC MGARCH | Dynamic Conditional Correlation Multivariate GARCH          |
| DE         | Diversification Effect                                      |
| DDOF       | Degree of Freedom                                           |
| DOW        | Dow Jones Industrial Average                                |
| E-GARCH    | Exponential GARCH                                           |
| ES         | Expected Shortfall                                          |
| FFF        | Flexible Fourier Functional                                 |
| GARCH      | Generalized Autor Regressive Conditional Heteroskedasticity |
| GIL        | Global interpreter lock                                     |
| IID        | Independent Identically Distributed Random Variables        |
| MCM        | Monte Carlo Method                                          |
| ML         | Maximum-Likelihood                                          |
| OO         | Object-Orientated                                           |
| P-GARCH    | Periodic GARCH                                              |
| P/L        | Profit and Loss                                             |
| RV-GARCH   | Realized Volatility GARCH                                   |
| SDLC       | Software Development Life Cycle                             |
| UI         | User Interface                                              |
| Var-Covar  | Variance-Covariance                                         |
| VaR        | Value At Risk                                               |
| VIX        | CBOE Volatility Index                                       |
| VWAP       | Volume Weighted Average Price                               |

## List of Figures

|                                                                                                                                                                                         |    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Figure 1:</b> SDLC model according to Langer (2016). Grey phases are not relevant in the scope of the thesis.....                                                                    | 19 |
| <b>Figure 2:</b> Flowchart of the app's processes.....                                                                                                                                  | 22 |
| <b>Figure 3:</b> De-seasonalizing Function according to Taylor and Xu (1997).....                                                                                                       | 26 |
| <b>Figure 4:</b> Outcome of de-seasonalizing returns of a day in the sample.....                                                                                                        | 27 |
| <b>Figure 5:</b> De-seasonalized return series across sample space. ....                                                                                                                | 28 |
| <b>Figure 6:</b> Univariate GARCH(1,1) computation according to Contino (2021).....                                                                                                     | 29 |
| <b>Figure 7:</b> Function transforming t-distributed standardized residuals to uniformly distributed residuals according to Contino (2021).....                                         | 29 |
| <b>Figure 8:</b> Log likelihood optimization of Gaussian Copula according to Contino (2021).....                                                                                        | 30 |
| <b>Figure 9:</b> DCC equation implementation according to Contino (2021).....                                                                                                           | 31 |
| <b>Figure 10:</b> Variance-Covariance Matrix forecast implementation. ....                                                                                                              | 31 |
| <b>Figure 11:</b> Step-ahead MCM VaR & ES estimation script. ....                                                                                                                       | 34 |
| <b>Figure 12:</b> Diversification effect function. ....                                                                                                                                 | 36 |
| <b>Figure 13:</b> 30-day ACF plots of research sample. ....                                                                                                                             | 56 |
| <b>Figure 14:</b> Subplot 1 - raw prices and smoothed VWAP prices, Subplot 2 – 1h volume, Subplot 3 - example VWAP return series, Subplot 4 - VWAP return series of entire sample. .... | 57 |
| <b>Figure 15:</b> VaR Exceedance occurrences across backtest window. ....                                                                                                               | 58 |
| <b>Figure 16:</b> App Tab 1 – (1/3).....                                                                                                                                                | 59 |
| <b>Figure 17:</b> App Tab 1 - (2/3). ....                                                                                                                                               | 60 |
| <b>Figure 18:</b> App Tab 1 - (3/3). ....                                                                                                                                               | 61 |
| <b>Figure 19:</b> App Tab 2. ....                                                                                                                                                       | 62 |
| <b>Figure 20:</b> Flowchart of App from Dash debugger - process from left to right. ....                                                                                                | 63 |

## List of Tables

|                                                                                                                                                                          |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Table 1:</b> Overview of multivariate ARCH-type modelling according to Bauwens et al. (2006).<br>.....                                                                | 7  |
| <b>Table 2:</b> DOW's top three percentage gains and losses. S&P Global 2018 cited in (Everything Explained Today 2021) .....                                            | 18 |
| <b>Table 3:</b> Price Sample Curation.....                                                                                                                               | 25 |
| <b>Table 4:</b> Backtest scenarios. ....                                                                                                                                 | 32 |
| <b>Table 5:</b> Backtested exceedances for $\text{VaR}_{\alpha=0.05}$ with different input days during the calm market scenario with normally distributed MCM draws..... | 33 |
| <b>Table 6:</b> Kupiec Backtest Result.....                                                                                                                              | 38 |
| <b>Table 7:</b> Backtest Output for the calm scenario using t-distributed random MCM draws.....                                                                          | 57 |

## 1. Introduction

On May 6<sup>th</sup> 2010, the Dow Jones Industrial Average (DOW) fell close to 10% and shortly recovered thereafter all in a matter of 30 minutes. An event later dubbed as the 2010 Flash Crash and commonly attributed to high-frequency trading algorithms. Individual securities were hit even harder with stocks such as Accenture trading below a dollar cent ([Bayer et al. 2010](#)). This was not a unique event as similar periods of sharp price corrections followed. Daily risk estimates based on daily returns, as commonly required by regulators, treat a day like the Flash Crash like every other as end of day returns are normal. For market participants with long-term horizons, this may indeed be the case. However, market participants with shorter intra-day time horizons such as market-makers, option-dealers or market supervisors would face detrimental false estimates of risk. From this example, it becomes apparent that intra-day time series data of security prices provides crucial information for estimating risk at lower time horizons. Estimating and monitoring such risk on intra-day time-series is the focus of this thesis.

Although academic research on time-series analysis goes back to the beginning of the 1920s, significant progress on financial time-series occurred later in the 1980s with the development of non-linear generalisation type models such as ARCH (autoregressive conditional heteroskedasticity) ([Zoubir 2017](#)). Further advancements in research have led to an extensive family of ARCH models, both in the univariate and multivariate form. A popular extension of univariate models is the GARCH (generalized) ARCH model that introduces a flexible lag parameter. Multivariate models enable modelling dependencies between securities. [Engle \(2002\)](#) developed the Dynamic Conditional Correlation (DCC) Multivariate (M) GARCH model that enables estimating non-linear correlations through time. In estimating Value At Risk (VaR) forecasts, the DCC MGARCH model has found to provide strong forecasting abilities on intraday data ([Morimoto and Kawasaki 2008](#)).

Although publications on practical implementations of different ARCH-type models exist, a software solution that implements the DCC MGARCH model to estimate risk was not found to the knowledge of the researcher. A potential reason is that these tools are developed by market participants without incentive for distribution to protect their competitive advantage. Therefore, the research aims to conceptualize a risk monitoring tool that utilizes the DCC MGARCH model for intraday risk management. The use case is designed around a simulation of a market-making type portfolio overseen by risk practitioners. The model however is also suitable for algorithmic trading. The research involves consideration from subtopics that include aspects of time-series analysis, risk and computer science.

The programme is written in Dash<sup>1</sup> and Python. Dash is a framework that enables writing both front-end and back-end code in pure Python. The researcher came to consider Dash due to related financial applications hosted on the platform's website ([Plotly 2022b](#)).

The research approach includes drafting a concept for the layout of the app that includes two tabs: A main tab for real-time monitoring and another to estimate risk towards the end of the day. The main DCC MGARCH model implementation is sourced from [Contino<sup>2</sup> \(2021\)](#) with the following specifications: (1) univariate standard GARCH(1,1) with (2) t-distributed errors with a (3) Gaussian copula for multivariate dependence. The script is adapted for implementation on intraday data at the 5-min frequency for which seasonality is removed by a seasonality term as defined by [Taylor and Xu \(1997\)<sup>3</sup> \(13\)](#). It is also extended to derive the forecasted variance-covariance matrix required for the forecasted VaR, estimated with the Monte Carlo Method (MCM) that includes t-distributed draws. The degree of freedom parameter (DDOF) is specified as the weighted average DDOF of the t-distribution of the univariate GARCH fits of the portfolio's securities. The sample consists of a series of 5-min returns of the Dow's index constituents from 09.12.2019 to 15.10.2021. The performance of the step ahead VaR forecasts is evaluated with a backtest over four different 5-day subsample periods. This includes one calm market phase and three in which volatility increases over the course of the backtesting window. The statistical significance is evaluated with a Kupiec Test.

Statistical significance at the 95% confidence level was achieved in three out of four subsample periods. Visual inspections of VaR exceedances in Appendix A Figure 15 indicate that apart from one scenario, they appear to be independently distributed across the backtesting window. Therefore, the current specifications show promising results for risk forecasting. Furthermore, the Dash framework provides a practical solution in abstracting away complexity involved in full stack web development and can enable risk practitioners, already familiar with supported data science languages like python to implement their own risk monitoring software. As a result, Dash could provide cost reduction potential for organizations if outsourcing can be omitted.

The thesis is structured as follows: section 2 defines the scope of research in further detail. Section 3 provides the theoretical background on the ARCH and GARCH process, multivariate GARCH models and the DCC model. Furthermore, the specificities of intraday returns and the

---

<sup>1</sup> Dash is open-source and licensed under the MIT License (MIT).

<sup>2</sup> Contino's python notebooks are licensed under the MIT License (MIT).

<sup>3</sup> An attempt to de-seasonalize returns by using volume weighted average prices (VWAP) was also made without success.



VaR are explained. Section 4 explains the concept of the programme structure. Section 5 explains the implementation. The implementation is evaluated in section 6. Section 7 and 8 discuss limitations and ideas for further research. A conclusion is provided in section 9. Two appendices are also provided for complementary resources.

Throughout the thesis, the terms ‘programme’, ‘app’, ‘risk monitoring tool’ and ‘script’ are used interchangeably. The latter is occasionally used to describe a subprocess but is understandable in context.

## **2. Scope of research**

As previously mentioned, the research topic is multifaceted, and subtopics fall under three broader categories. On behalf of time series analysis, (1) modelling volatility and (2) studying intraday stylized facts. On behalf of risk management, (3) VaR estimation. Some elements of computer science are also required although an economic focus is maintained throughout the thesis.

Therefore, the topic requires some restrictions. Regarding intraday-seasonality, only two approaches are considered, implementing a seasonality term and VWAP. The former is implemented<sup>4</sup> in its basic form considering 5-minute seasonality and ignoring broader seasonality’s such as weekday effects or volatility around scheduled news events.

From different type of multivariate GARCH models, the DCC variant has been studied and is implemented with one specific model calibration. A standard univariate GARCH(1,1) with a Gaussian copula for joint dependence. Resulting VaR estimation necessary for model evaluation is performed with the MCM. Risk is considered for internal purposes and not on regulatory considerations. The programme and model evaluation is currently restricted to the 30 securities over a 2-year sample size.

---

<sup>4</sup> The VWAP approach did not achieve the desired outcome. Further explanation provided in section 3.2 and 5.2.

### 3. Theoretical background

Section 3.1 provides an overview of the GARCH process in the univariate case and explains how the model extends to the multivariate setting. Issues related to the incorporation of intraday data are discussed in section 3.2. Then, both the univariate and multivariate model evaluation is explained in section 3.3. After clarifying the model's specifications and seasonality, section 3.4 discusses modelling risk using VaR. Finally, section 3.5 incorporates a historical simulation as a further risk estimate for potential losses.

#### 3.1. GARCH models

First, the univariate ARCH and GARCH model is explained. This is necessary as the modelled variances are used as input for the class of multivariate GARCH models, explained thereafter. Finally, the model of choice, DCC MGARCH, is explained.

##### 3.1.1. ARCH process

The univariate GARCH model was developed as an extension of the ARCH (Autoregressive Conditional Heteroskedastic) model introduced by Engle in 1982 (Bollerslev 1986). The use of ordinary least squared (OLS) linear regression is commonly used in econometrics to study the relationship between dependant and independent variables along a given time-series. A key characteristic of this approach is the assumption of homoskedasticity, that magnitude of square errors is assumed to be constant across all observations. However, empirical observations of volatility show patterns of serial correlation. That is, periods of high (low) volatility tend to be followed by periods of high (low) volatility.<sup>5</sup> Campbell et al. (2012, p.481) conclude that the use of volatility measures based around the assumption of constant volatility is both statistically inefficient and logically inconsistent. Engle (1982) introduced the ARCH model to account for this stylized fact. The conditional variance for an ARCH (1) model is written as a distributed lag of past squared errors (innovations):

$$\sigma_t^2 = w + \alpha_1 r_{t-1}^2 \quad (1)$$

where  $w$  is a white noise stochastic process,  $\alpha_1$  is a constant and  $r_{t-1}^2$  is the squared error term of  $t-1$ .  $\alpha_1 > 0$  so that conditional variance remains positive. ARCH( $p$ ) models incorporate  $p$  lags. As  $p$  increases, the persistence in the modelled variance increases. Bera and Higgins (1993) attribute the success of the model to its simplicity and ability to capture several observed features of financial data such as fat-tailed distributions, clustering effects, nonlinearity as well

---

<sup>5</sup> This observation also holds for the sample (Figure 5).

as changes in the ability to forecast future values. The model is not restrained to volatility data but has proven effective in the study of macroeconomic time series and risk premia (Bollerslev 1986). Aspects of testing for the fit of the model are discussed in section 5.4.

### 3.1.2. GARCH process

According to Bollerslev (1986), the ARCH model relies on an arbitrary linear declining lag structure that fails to account for a longer term memory effect evident from empirical work on volatility modelling. He therefore introduced the GARCH (Generalized Autoregressive Conditional Heteroskedastic) model that allows for an increased flexibility in the lag structure. The GARCH (1,1) model may be expressed as follows:

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad (2)$$

$$r_t = \sigma_t Z_t \quad (3)$$

where  $\alpha_0$ ,  $\alpha_1$  and  $\beta_1$  denote the estimated model parameters with constraints  $\alpha_0 > 0$ ,  $\alpha_1 \geq 0$ ,  $\beta_1 \geq 0$ .  $r_{t-1}^2$  is the lagged squared error term<sup>6</sup> and  $\sigma_{t-1}^2$  the lagged conditional variance. The latter term allows for the flexible lag structure. In the base model, the error term,  $r_t$  depends on independent identically (IID) (0,1) standard normally distributed innovations,  $Z_t$  scaled by the current volatility  $\sigma_t$ . As financial returns typically distributed with fatter tails, t-distributed errors are used in the programme. The optimum parameters are obtained from maximum-likelihood (ML) estimation. An algorithm, typically Newton-Raphson, is used to incrementally test optimum parameters along the time-series for a given distribution. Initially, a starting variance must be estimated, commonly the observed variance of the residuals. Then the first squared residual and the first variance are used to estimate the variance of the next term and the process continues for the complete time series (Engle 2001). In python, the ARCH package (Sheppard 2021) already provides this. It implements log-likelihood as the default estimator. The formula is like (12).

In this thesis, the t-distribution is used due to the lower probabilities around the shoulders of the bell curve and higher probabilities at the tail end of the distribution. Further common choices are the skewed t-distribution, the normal distribution or the pareto distribution common from extreme value theory can also be applied. The optimum distribution itself may be subject to analysis and testing<sup>7</sup> but is not further tested in the scope of this research.

---

<sup>6</sup> Present from the ARCH model.

<sup>7</sup> Visualization tests such as histograms or QQ-plots or statistical tests such as the Shapiro-Wilk test for normal distributions or the Anderson-Darling- and Kolmogorov-Smirnov test for choice of other distributions.

Another choice the researcher faces is whether to opt for a GARCH(1,1) or a GARCH(p,q) model. The latter introduces further complexity and computational expense for (1) the univariate GARCH parametrization and (2) for forecasting the conditional variance-covariance matrix for  $t+1$  as further recursive functions for the lags are required. Empirical work by [Hansen and Lunde \(2005\)](#) of over 300 ARCH type models on exchange rate data and IBM stock returns has found no evidence that the GARCH(1,1) is outperformed by more sophisticated models. Therefore, the programme implemented in this research uses the GARCH(1,1) model.

Specifying the GARCH parameters is the first step and serves as an input for estimating the DCC parameters of the DCC MGARCH model explained in section 3.1.4.

### 3.1.3. Multivariate GARCH

Univariate ARCH models have experienced broad adoption in forecasting volatility since the published seminal paper by [Engle \(1982\)](#). It is widely accepted that volatilities of securities or markets are not independent but move together. Univariate models fail to account for this. MGARCH models, however, can model the dynamics of covariance and correlations between asset returns and thus enable, for instance, the study of volatility spill overs between markets or long-run development of volatilities in the global market under the influence of globalisation. Further applications of the models are the following: specification of time-varying hedge ratios or modelling time-varying  $\beta$  coefficients of the market return factor in the CAPM model ([Bauwens et al. 2006](#)).

[Bauwens et al. \(2006\)](#) provide an overview and classify the family of MGARCH models as summarized in Table 1.<sup>8</sup>

When generalizing the univariate GARCH model to the multivariate case, the error term vector,  $\epsilon_t$ , of a  $d$ - dimensional time series model assumes a conditional mean of zero and a conditional covariance matrix given by the positive definitive ( $d \times d$ ) matrix  $\mathbf{H}_t$ :

$$\epsilon_t = \mathbf{H}_t^{\frac{1}{2}} \xi_t \quad (4)$$

where the innovation vector,  $\xi_t$ , is IID with mean zero and the covariance matrix equals the identity matrix  $\mathbf{I}_d$ . Like the univariate case of a GARCH(1,1) model,  $\mathbf{H}_t$  depends on the lagged error term vector  $\epsilon_{t-1}$  and the lagged conditional covariance matrix  $\mathbf{H}_{t-1}$ . As the dimensions of the vector increases, so do the dimension of the required parameter space ([Humboldt University](#)

---

<sup>8</sup> The MGARCH model survey by [Almeida et al. \(2018\)](#) also provides a great overview.

Berlin, School of Business and Economics n.D.). Redundant relations are parametrized due to the symmetry of the variance-covariance matrix and the cross product of mean residuals (Billio et al. 2003). The first family of models shown in Table 1 aim to reduce this complexity in parameters with restrictions and additional structures.<sup>9</sup> The second class of models are models that allow for linear combination of a variety of univariate models. For instance, EGARCH or APARCH. The third category entails models that can be viewed as nonlinear combinations of univariate GARCH models. This enables two customizations. First, the separate choice of univariate individual conditional variance as with the previous class. Second, modelling dependence of the conditional correlation matrix, for instance with the joint density function of a copula.

Another benefit of this third class is they require less parameters than the first category models and are easier to estimate (Bauwens et al. 2006).

**Table 1:** Overview of multivariate ARCH-type modelling according to Bauwens et al. (2006).

| Category                                               | Type       | Models                                                                                                              |
|--------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------|
| Generalizations of the Univariate Standard GARCH Model | Linear     | Vector Error Correction (VEC) <sup>10</sup><br>BEKK<br>Factor Models                                                |
| Linear Combinations of Univariate GARCH Models         | Linear     | Orthogonal models or latent factor models with univariate models:<br>O-GARCH<br>GO-GARCH                            |
| Nonlinear Combinations of Univariate GARCH Models      | Non-Linear | Constant Conditional Correlation (CCC)<br>Dynamic Conditional Correlation (DCC)<br>General Dynamic Covariance (GDC) |
| Alternative Approaches                                 | -          | Stochastic Volatility (SV) models<br>Realized Volatility                                                            |

<sup>9</sup> Such as factors or diagonal parameter matrices.

<sup>10</sup> Also referred to as VEC model in the literature.

### 3.1.4. DCC multivariate GARCH

Due to the scope of the thesis, the programme is reduced to incorporate one of the MGARCH models although it could be extended to enable choice for different models. [Morimoto and Kawasaki \(2008\)](#) review five models and compare their performance in forecasting intraday VaR. The reviewed models were (1) VEC model (2) BEKK model (3) diagonal model<sup>11</sup> (4) CCC (5) DCC. They find the DCC to yield the best results and consider it favourable for practical risk management.

It was hence chosen in this research and is discussed in more detail. The DCC model was developed by both [Engle \(2002\)](#) and [Tse and Tsui \(2002\)](#) in the same year with some differences. The former is implemented in this research. The conditional covariance matrix Matrix,  $\mathbf{H}_t$ , is given by:

$$\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t, \quad (5)$$

where the diagonal matrix of conditional standard deviation,  $\mathbf{D}_t$ , is

$$\mathbf{D}_t = \text{diag} \left( h_{11,t}^{\frac{1}{2}} \dots h_{NN,t}^{\frac{1}{2}} \right) \quad (6)$$

and the dynamic correlation matrix,  $\mathbf{R}_t$ , is expressed as:

$$\mathbf{R}_t = \text{diag} \left( q_{11,t}^{-\frac{1}{2}} \dots q_{NN,t}^{-\frac{1}{2}} \right) \mathbf{Q}_t \text{diag} \left( q_{11,t}^{-\frac{1}{2}} \dots q_{NN,t}^{-\frac{1}{2}} \right) \quad (7)$$

$\mathbf{Q}_t$  is a  $N \times N$  symmetric positive definitive matrix expressed as:

$$\mathbf{Q}_t = (1 - \alpha - \beta) \bar{\mathbf{Q}} + \alpha \boldsymbol{\epsilon}_{t-1} \boldsymbol{\epsilon}_{t-1}^T + \beta \mathbf{Q}_{t-1} \quad (8)$$

where parameters  $\alpha$  and  $\beta$  are non-negative and satisfy  $\alpha + \beta < 1$ .  $\bar{\mathbf{Q}}$  is the long run unconditional variance covariance matrix.

The researcher faces the decision on how to model the error vector<sup>12</sup>,  $\boldsymbol{\epsilon}_t$ . One approach is to implement Copulas to model joint dependence.

<sup>11</sup> The authors define it as a model that reduces parameters of a multivariate GARCH model by estimating parameters from each univariate GARCH model independently. It is not classified in the [Bauwens et al. \(2006\)](#) paper to the authors knowledge.

<sup>12</sup> This is also referred to as the immediate disturbances vector throughout the paper and the written code.

According to Sklar's (1959) theorem, the connection between a joint distribution of an  $N$ -dimensional distribution function  $\mathbf{F}$  and marginal distributions  $F_i, i = 1, \dots, N$  of unconditional distributions  $x_i, i = 1, \dots, N$  with a copula  $\mathbf{C}$  can be formulated as:

$$\mathbf{F}(x_1, \dots, x_n) = \mathbf{C}(F_1(x_1), \dots, F_n(x_n)) \quad (9)$$

where  $\mathbf{C}$  is a cumulative distribution function. Contino's (2021) DCC model script was utilized in the programme and the choice of modelling dependence was not changed. Contino (2021) uses a standard GARCH(1,1) model with t-distributed errors. The t-distributed standardised errors of each variable are recovered and transformed to a uniform distribution on the interval  $[0,1]$  via implementation of a cumulative distribution function (CDF). This is a prerequisite for copulas such that the copula density takes values between  $[0,1]$ . The joint dependence of the univariate uniform distributor is then modelled with the use of a gaussian (normal) copula.

An  $N$  dimensional Gaussian copula is defined as:

$$\mathbf{C}(u_{1,t}, \dots, u_{N,t}) = \eta(\Phi^{-1}(u_{1,t}), \dots, \Phi^{-1}(u_{N,t}) | \mathbf{0}, \mathbf{R}_t) \quad (10)$$

where  $u_i = F_i(x_i)$ ,  $\eta$  is the multivariate normal CDF,  $\Phi^{-1}$  is the univariate normal inverse CDF.

The copula density is then given by:

$$\mathbf{c}(u_{1,t}, \dots, u_{N,t}) = |\mathbf{R}_t|^{-\frac{1}{2}} \exp \left[ -\frac{1}{2} \mathbf{x}_t' (\mathbf{R}_t^{-1} - \mathbf{I}_N) \mathbf{x}_t \right] \quad (11)$$

with correlation matrix  $\mathbf{R}_t$ ,  $N$  dimensional identity matrix  $\mathbf{I}_N$  and  $\mathbf{x}_t = (x_{1,t}, \dots, x_{N,t})$  (Contino 2016, p.69).  $\mathbf{x}_t$  is required for the ML estimation in equation (12).

Parameters  $\alpha$  and  $\beta$  can be estimated via ML estimation like in the univariate GARCH process.

The log likelihood for a normal copula is defined as:

$$\log L(\mathbf{R} | u_t) = \frac{1}{2} \sum_{t=1}^T (\ln |\mathbf{R}| + \mathbf{x}_t' (\mathbf{R}^{-1} - \mathbf{I}) \mathbf{x}_t) \quad (12)$$

where  $\mathbf{x}_t = (x_{1,t}, \dots, x_{N,t})$  and  $x_{i,t} = \Phi^{-1}(u_{i,t})$  (Contino 2016, p. 106). The implementation is discussed in section 5.3.

Contino's (2021) script has been extended to incorporate a forecasted variance-covariance matrix,  $\mathbf{H}_{t+1}$  for  $t + 5$  min. To derive the forecasted matrix, (8) that calculates  $\mathbf{Q}_t$  needs to be changed such that  $\epsilon_{t-1} \epsilon_{t-1}^T$  is a matrix of zeros and  $\mathbf{Q}_{t-1}$  is the last  $\mathbf{Q}_t$  obtained from the DCC MGARCH process. Parameter values of  $\alpha$  and  $\beta$  are also obtained from this previous process.

The further the forecasting horizon, the more the modelled variance-covariance matrix reverts to the long-term variance-covariance matrix,  $\bar{\mathbf{Q}}$ . Furthermore, the univariate variances from the GARCH process needs to be forecasted to  $t+1$  as well to compute  $\mathbf{D}_t$  correctly in the model.

A further modelling choice is how much historical data is used as input for the model. There is a potential trade-off between accuracy and performance. What could drive this decision is knowledge about the persistence of volatility shocks to decide on sample length. Unfortunately, there is no consensus economic model to capture conditional variance adequately (Diebold and Lopez 1995).

However, Ng and LAM (2006) have studied the effects of sample size on the standard GARCH(1,1) model using daily sample data. They conclude that a sample size of 1000 datapoints is adequate. The authors remark that it is subject to further research what requirements are needed for high frequency intraday data. How this translate to MGARCH models is also not specified. Therefore, different input sizes are tested for the model.<sup>13</sup> The results are discussed in Section 6. The choice of sample size also affects the de-seasonalizing of intraday returns. The greater the sample size, the greater the average of the seasonality term. See (13). With growing sample size, the computation time for the de-seasonalizing process will also increase.

After considering the DCC MGARCH model specifications, decisions on how to handle high-frequency intraday sample data are described in section 3.2 below.

### 3.2. Intraday Seasonality

The DCC MGARCH model was originally developed for implementation on daily time series (Engle 2002). Although handling daily data is convenient and inexpensive, it is not as informative as data at higher frequencies (Będowska-Sójka 2015). This becomes obvious in the occasions of flash crashes mentioned in section 1.

However, intraday data provides additional challenges in time-series modelling as it is subject to unique stylized facts. Volatility of returns are typically elevated towards the opening and close phase of a trading day and lower around noon. Both Harris (1986) and Wood et al. (1985) have been one of the first to detect and study U-shaped patterns in the volatility of returns linked to the scale of new information available to the market for price action and the return generating process. Bollerslev and Andersen (1997) and Engle (2002) demonstrate, that fitting traditional

---

<sup>13</sup> See Table 5.



GARCH models on intraday return series introduces a bias if the periodicity in the volatility process is not respected. To overcome this shortfall, the DCC MGARCH model requires de-seasonalized returns as input.

There are multiple approaches to address seasonality in the literature. [Bollerslev and Andersen \(1997\)](#) propose a flexible fourier function (FFF) to address seasonality by modelling frequencies. [Bollerslev and Ghysels \(1996\)](#) introduce a periodic(P-) GARCH to capture the repetitive time variation in variance via an additional dummy variable. Further methods exist and are well summarized by [Boudt et al. \(2008\)](#).

[Taylor and Xu \(1997\)](#) propose a non-parametric approach that introduces a seasonality term by averaging out variance across intraday periods. [Martens et al. \(2002\)](#) have reviewed these methods on thirty-minute exchange rate data of the Deutsche Mark and Japanese Yen versus the US Dollar and conclude their performance in increasing order as follows: simple seasonality term, log seasonality term, FFF and P-GARCH. However, the P-GARCH performed only marginally better albeit a higher computational expense.

The programme script developed in the scope of this thesis follows the simple seasonality approach and is covered in more detail. [Karmakar and Paul \(2019\)](#) describe [Taylor and Xu's \(1997\)](#) approach as below.

De-seasonalized returns,  $rd_{t,n}$ , are defined as follows:

$$rd_{t,n} = \frac{r_{t,n}}{S_{t,n}}, n = 1, 2, \dots, N, \quad (13)$$

where  $r_{t,n}$  is the nth intraday return at day t and  $S_{t,n}$  is the seasonality term for N intraday periods. For  $S_{t,n}$ , the squared returns are averages for each intraday period:

$$S^2_{t,n} = \frac{1}{T} \sum_{t=1}^T r_{t,n}^2, n = 1, 2, \dots, N, \quad (14)$$

where T is the total number of days in the sample.

Empirical research has revealed the existence of a wide variety of further stylised facts of volatility in intraday periods. These include changes in kurtosis and correlations throughout the trading day, weekday periodicity or volatility around scheduled news events. These are discussed in more detail in section 7.1.4.

Volatility is not the only metric in markets to experience periodicity. Bińkowski and Lehalle (2018) show periodicity in the metrics of ‘traded quantity’ or volume, the average bid-ask spread as well as the average size at first volume. Notably, volume experiences a similar “U” shaped periodicity with the highest levels on market open, followed by a flat period during the day and an elevation towards market end. The authors stress that the periodicity is less profound than that experienced by volatility. Incorporation of the volume metric in de-seasonalizing data has not been found in the process of the literature review. Both this and the suggestion of Dr. Christian Contino (personal communication, November 30, 2021) to attempt to incorporate this metric to address volatility periodicity led to the consideration of incorporating returns derived by volume weighted average prices (VWAP).

With the price series  $(P_i)_{i=1,\dots,N}$  and the coherent volumes  $(V_i)_{i=1,\dots,N}$ , VWAP is defined as

$$\text{VWAP} = \frac{\sum_{i=1}^N V_i P_i}{\sum_{i=1}^N V_i} \quad (15)$$

The price and volume series used in this research are defined by a 1 hour-rolling series consisting of 12 datapoints at 5-min frequencies. As VWAP is a daily indicator, it on every trading day. Therefore, the VWAP progresses into a rolling series during the first hour of trading with  $(P_i)_{i=1}, (V_i)_{i=1}$  followed by  $(P_i)_{i=1,2}, (V_i)_{i=1,2}$  up to  $(P_i)_{i=1,\dots,12}, (V_i)_{i=1,\dots,12}$ .

This approach was not implemented in the final script as it fails to remove intraday seasonality. This is further explained in section 5.2. See also Appendix A Figure 14.

### 3.3. Model Evaluation

After considering the specifications for the model building process, it is important to consider evaluating the validity of the model. The goodness of fit can be assessed on both the univariate and multivariate level.

#### 3.3.1. Univariate fit

Several methods for validating univariate fit exist. This includes plotting the GARCH error term,  $Z_t$ , and visually inspecting whether the error terms are random and IID. In addition, an autocorrelation function (ACF) can be used to inspect autocorrelation at a certain confidence

interval, commonly 95%. This function is restricted to testing for randomness at each interval in the time series. Therefore, the Ljung-Box test can be applied to test for randomness over several lags. Another common approach is to visually inspect errors with a Q-Q plot. The quantile of the assumed distribution is plotted against the empirical quantiles of the time series. The plot should approach linearity as an indication that the choice of distribution is correct (Orskaug 2009, p. 33-34). Although it is important to consider the univariate fit, it has not yet been included in the programme. However, the GARCH parameters along with their coinciding p-values are displayed to enable an initial validation whether univariate GARCH parameters are plausible. This is discussed further in section 5.4.

### 3.3.2. Multivariate fit

The univariate fit is followed by consideration of the multivariate fit. Even if the marginals from the univariate model do fit, the multivariate distribution (the gaussian copula), does not have to fit. Therefore, it is important to validate the multivariate fit (Orskaug 2009, p. 34) The author notes that literature on testing multivariate fit is scarce. The author describes both the in-sample Baringhaus-Franz multivariate test and the out-of-sample Kupiec and Christoffersen test.

In this research, the out-of-sample Kupiec test is implemented. Initially, the mentioned out-of-sample tests require backtesting VaR<sup>14</sup>. The intuition behind the approach is to define a historical period in which VaR forecasts are compared to realised profit and losses (P/L). Exceedances of the VaR are traced and should not occur more often than at the defined  $\alpha$  level. Formally, this is defined as:

$$\mathbf{I}_{\alpha,t} = \begin{cases} 0 & \text{if loss } p^{t-1} - p^t \text{ is less than or equal to VaR} \\ 1 & \text{if loss } p^{t-1} - p^t \text{ exceeds VaR} \end{cases} \quad (16)$$

where  $\mathbf{I}$  is the univariate exceedance process, and  $p$  is the portfolio value. The process is expected to follow an IID Bernoulli process with the success probability of  $\alpha$ . It is important to differentiate between model testing for regulatory- or internal purposes. Under Basel IV, a traffic light table for exceedances is defined for backtesting of 250 trading days. Proprietary VaR models are rejected if they fall into the red zone in excess of 10 exceedances. However, this approach does not follow a statistical hypothesis test of an observed exceedance process

---

<sup>14</sup> VaR is explained in Section 3.4.

but is generated from a binomial distribution with the null hypothesis that exceedances do not surpass the VaR model at  $\alpha=0.01$  (Holton 2022).

For internal purposes however, more advanced techniques may be applied. In contrast to comparing exceedances to a standardised regulatory table, the Kupiec test studies the binomial process of the observed exceedances and forms a test statistic.

As mentioned, the exceedances follow a binomial distribution:

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad (17)$$

where  $n$  is the sample length,  $x$  the exceedances and  $p$  is the probability of a violation. The exceedances increase pro rata with sample length.

The Kupiec test defines the following set of hypotheses:

$H_0$ : The expected proportion of exceedances is equal to  $\alpha$ .

$H_1$ : The expected proportion of exceedances is not equal to  $\alpha$ .

The test statistic of the null hypothesis is then defined as:

$$\begin{aligned} \text{Kupiec} &= 2 \ln \left\{ \frac{\binom{n}{x} p_{\text{obs}}^x (1-p_{\text{obs}})^{n-x}}{\binom{n}{x} a^x (1-a)^{n-x}} \right\} \\ &= 2 \ln \left[ \left( \frac{x}{n} \right)^x \left( 1 - \frac{x}{n} \right)^{n-x} \right] - 2 \ln [a^x (1-a)^{n-x}] \sim \chi^2(1 \text{ dof}), \text{ as } n \rightarrow \infty \end{aligned} \quad (18)$$

where  $\left( \frac{x}{n} \right) = p_{\text{obs}}^x$ , is the observed proportion of exceedances.

The test statistic follows a chi-squared asymptotic distribution with 1 degree of freedom. Critical values can thus be derived from chi-squared tables. If  $p_{\text{obs}}^x$  is greater than the critical values at the significance level  $\alpha$ , the model is accepted. The Kupiec test is also referred to as an unconditional coverage test, as the test does not account for when the exceedances take place. To examine whether the exceedances are IID across time, the Christoffersen test can be applied (Orskaug 2009, p. 36) In the scope of this research, a simple plot<sup>15</sup> of exceedances was performed to inspect for potential instances of clusters.

---

<sup>15</sup> See Appendix A Figure 15.

### 3.4. Value At Risk

With both the derived step ahead forecasted multivariate volatility and scrutiny via verification of model fit, it is now possible to derive forward looking market risk measures. One of the broadly adopted methodologies for defining risk is the Value At Risk (VaR) model. The goal is to determine a loss level for a portfolio at a predefined confidence interval, dependant on a distribution. It has become an industry standard for regulation.<sup>16</sup> It is important to note that the motivation for estimating risk figures in this thesis was not under the influence of regulatory requirements but rather to give practitioners further guidance to manage and exploit risk.

There are different approaches to calculating VaR. These are broadly classified as parametric- or non- (half-) parametric. The former includes the variance-covariance method. The latter includes the use of a Monte Carlo simulation. For the variance-covariance method, VaR is expressed as:

$$\text{VaR}_{\Delta t, \alpha} = V(-N_{1-\alpha} \sigma - \mu) \quad (19)$$

where  $\sigma$  is the portfolio standard deviation,  $\mu$  portfolio mean return,  $\alpha$  the confidence interval and  $V$  the present value of the portfolio.  $N_{1-\alpha}$  is the inverse of the standard normal cumulative distribution. Under a normal distribution, it is also referred to as the z-score, the number of standard deviations above or below the mean. A negative pre-sign ensures a positive VaR risk figure. This only holds for  $\alpha < 0.5$ . Common practice is to implement alpha values of 0.01 or 0.05 to capture risk at the tail end of the distribution. The central assumption is that both asset returns and the variance-covariance matrix,  $\mathbf{H}_t$ , are multivariate normal (Walther 2020):

$$\Delta V_{\Delta t} \sim N(\mu, \mathbf{H}_t) \quad (20)$$

The multivariate normal assumption is controversial however as correlations of assets converge to 1 in periods of market stress as the risk-off paradigm affects asset prices simultaneously (Chua et al. 2009).

---

<sup>16</sup> See requirements on disclosing market risk under the Basel accords.

Commonly, variances and covariances in  $\mathbf{H}_t$  are calculated as a linear function of historical data. For variance:

$$s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1} \quad (21)$$

The aim of the DCC MGARCH model is to capture the dependence between assets dynamically such that  $\mathbf{H}_t$  adapts in periods of market stress, providing a more accurate input for the VaR estimation. [Angelidis et al. \(2004\)](#) study VaR forecasting using GARCH models and support this view by finding that volatility forecasting accuracy increases with increased flexibility of the GARCH model.

As a complementary risk figure, the Expected Shortfall (ES or cVaR<sup>17</sup>) is often calculated alongside VaR to capture the average incurred losses in the tail end of the distribution exceeding VaR. ES is calculated as:

$$ES = V \left( -\mu + \frac{\sigma n(N_{1-\sigma})}{1-\sigma} \right) \quad (22)$$

where  $n$  is the probability density function.

The programme incorporates the MCM method due to the relative ease of accounting for exposure of long-short portfolios. The variance-covariance method requires additional considerations in this setting. With use of the conventional formula, exposures of long and short securities would net out, leading to an underestimated VaR ([Straja 2004](#)). The variance-covariance method is only implemented to determine the VaR of the individual assets in order to determine diversification effects.<sup>18</sup>

The MCM is used to simulate portfolio values for  $t+1$  to calculate a step ahead VaR on the first tab of the programme. On the second tab, ‘adverse scenario’ the MCM is implemented to simulate values up towards the end of the trading day to estimate an end-of day.

The MCM is a simulation based VaR approach that uses computer-generated correlated random returns,  $R$ , and uses the law of large numbers to infer estimates based on the distribution of  $R$  ([Packham 2021](#)). For the algorithm, generating  $k$  scenarios for an  $n$ -dimensional random vector

---

<sup>17</sup> Conditional Value at Risk.

<sup>18</sup> This is explained at the end of the section.

of assets,  $(\Delta Y_1, \dots, \Delta Y_d)$  follows a three-step process. First,  $k$  vectors are randomly drawn independently from a normal distribution  $(u_1, \dots, u_d)$ .

These random numbers are transformed to independent standard normal random numbers:

$$(x_1, \dots, x_d) = (N^{-1}(u_1), \dots, N^{-1}(u_d)) \quad (23)$$

such that  $\mu=0$  and  $\sigma=1$  is satisfied. Finally, the Cholesky-decomposition,  $\mathbf{B}$ , that satisfies

$$\text{Cov}(\Delta \mathbf{Y}) = \mathbf{B}\mathbf{B}^T \quad (24)$$

is used to transform the standard random numbers to correlated random numbers:

$$\Delta \mathbf{Y} = \mu + \mathbf{B}\mathbf{x}_i \quad (i=1, \dots, d) \quad (25)$$

where random vector  $\Delta \mathbf{Y}$  is now jointly normally distributed ([Walther 2020](#)).

Random correlated returns are then used to simulate  $k$  different portfolio developments for  $t+1$ . The process may be repeated for  $n$  time periods into the future. The VaR can then be estimated as:

$$\text{VaR}_{\Delta t, \alpha} = V_{\Delta t, \alpha} - V_0 \quad (26)$$

where the difference between the portfolio value at the percentile of confidence interval  $1 - \alpha$ , and the initial portfolio value,  $V_0$ , then serves as the VaR measure. Expected Shortfall is estimated as the difference of the average of simulated portfolios at a given point in time that lie beneath the confidence interval and  $V_0$ .

A potential benefit of the MCM is that the output can be presented graphically across time and provide the risk practitioner with an understanding of how the portfolio may evolve over time.

The above-mentioned approach faces a shortcoming in that the normal distribution assumption underestimates the tail-risk, common in financial time series. Consequently, the approach can be modified to draw random returns from a different distribution such as a t-distribution. The t-distribution introduces the degree of freedom parameter that needs to be calibrated. There are two approaches that include estimating DDOF on the portfolio level or on a per asset basis. The former requires ML estimation of the standardised residuals of portfolio returns. Alternatively, the distributions of standardised residuals of individual securities can be used as an estimate. The thesis follows the latter approach due to a subjectively easier mode of implementation

within the code. The DDOF parameter is therefore defined as the weighted average DDOF of the univariate GARCH model fits.

VaR also allows for the computation of the diversification effect (DE) under the assumption of sub-additivity. It is expressed as the aggregated isolated VaR of the individual assets minus the combined VaR:

$$DE = (VaR_1 + VaR_2) - VaR_{1,2} \quad (27)$$

This can be extended on a portfolio level by calculating DE's for each asset by comparing the VaR of the asset with the rest of the portfolio. For a portfolio of more than 2 assets, every asset will obtain unique diversification effects based upon its position size and dependence to other assets in the portfolio. This can be used as a further risk metric for trading decisions. See the respective column in the position table in Appendix B Figure 17.

### 3.5. Historical Scenario

An additional risk assessment was implemented for an additional comparison on how the portfolio value may evolve to the end-of day compared to the MCM. This does not use the DCC model approach and instead simulates how the portfolio's value evolves if each asset in the portfolio experiences the average historical 5-min returns from Table 2, dependant on the choice of the user in a dropdown menu in the programme. The user can see how the portfolio's MCM VaR & ES risk figure compares to a historical simulation type approach.

**Table 2:** DOW's top three percentage gains and losses. S&P Global 2018 cited in [\(Everything Explained Today 2021\)](#)

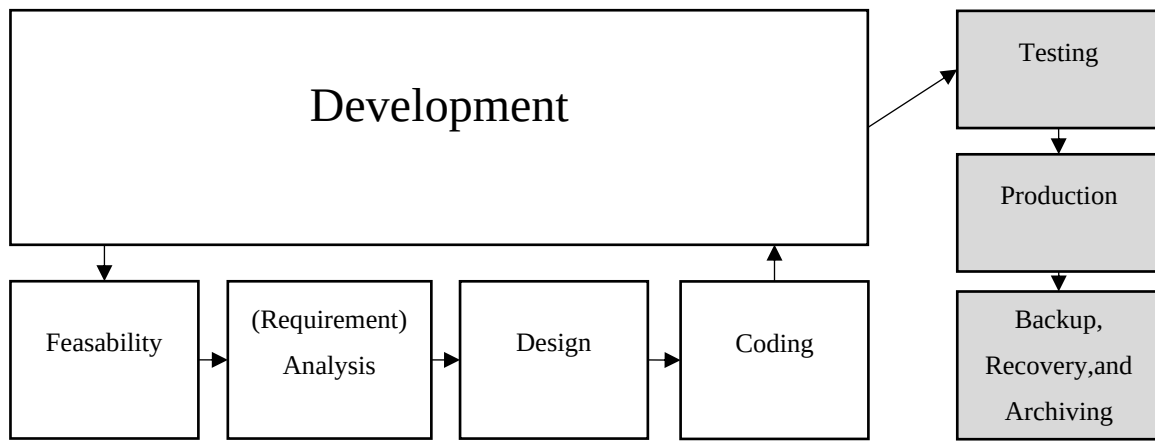
| Date       | Event                         | Daily Return | Average 5-min Return |
|------------|-------------------------------|--------------|----------------------|
| 1987-10-19 | Black Monday 1987             | -0.2261      | -0.00332             |
| 2020-03-16 | 2020 Stock Market Crash       | -0.1293      | -0.0018              |
| 1929-10-28 | Wall Street Crash of 1929     | -0.1282      | -0.00178             |
| 1933-03-15 | Emergency Banking Act 1933    | 0.1534       | 0.001855             |
| 1931-10-06 | Financial Bill By Hoover 1931 | 0.1487       | 0.001802             |
| 1929-10-06 | 1929 Pre Crash Rise           | 0.1234       | 0.001512             |



## 4. Concept and Design

This section of the thesis discusses the conceptualization of the application and its design. The considerations are structured according to the software development life cycle (SDLC) methodology ([Langer 2016](#)). There is not one model that practitioners follow by convention, a whole body of literature has evolved around adaptations of the model ranging from more rigid to agile models and further dissections of intermediate stages. In this chapter, relevant stages are described to provide guidance to the development process of the programme. Many steps are omitted due to the comparably small scale of the project and economic focus of the thesis.

### 4.1. Software Development Life Cycle



**Figure 1:** SDLC model according to [Langer \(2016\)](#). Grey phases are not relevant in the scope of the thesis.

The SDLC is a tool to manage the development of software across different phases ranging from assessing the feasibility of a project to production and maintenance. The analysis phase includes developers and stakeholders determining all the specifications and requirements of the product. These are formally defined and documented thereafter. This is followed by the design subphase in which the structure of the software is planned. This includes deciding on the programming language used and defining various objects in case of an object orientated (OO) approach. Designing the user interface (UI) is also included in this step. The coding subphase deals with decisions about the architecture and logic of the code, including decisions about programming paradigms such as structured versus coupled programming ([Langer 2016](#), p. 8). The testing, production and backup phase are not described here as the programme developed is a proof of concept without a current intent of deployment.

## 4.2. Requirement Analysis

The application is intended as a web-based application. The front-end UI consist of two separate tabs with unique features. The first tab is intended to simulate how the risk management could operate and be used by practitioners on an ongoing basis for risk monitoring. It therefore focuses on both the DCC MGARCH modelling and short-term risk, namely a 5-minute step-ahead VaR. This tab consists of a main Callback<sup>19</sup> that simulates a trading environment and models forecasted volatility and risk measures. Additional Callbacks give users additional features that can be triggered such as toggling visibility of data and risk metrics. The second tab is designed to show how the position develops to the end of day under certain scenarios. This can be thought of as a stress test but is named ‘adverse scenario analyses’ to avoid confusion with respect to regulatory definitions of term. The requirements of each tabs are listed in more detail below. Figure 2 provides a visual overview with a flowchart.

Tab 1:

- Simulation parameters. Give the user the choice to adjust the bid-ask spread, maximum order size of randomly generated orders, how many orders should be simulated for every iteration, the margin requirement on short positions and the date of simulation.
- Live market data script. For demonstrative purposes, a script needs to simulate the inflow of new market data on an iterative basis. This is dependent on the date the user chooses in the parameters. A real word application would source current market data from a data service provider. This was not feasible in the scope of this project due to the related costs of real time intraday data.
- Random order generator. A logic that simulates both random order requests and buy-in prices by simulating a bid-ask spread on current market prices. In a real world, this data would likely stem from a standalone trading software.
- Portfolio position updater. This script tracks both new positions and existing positions with their relative buy-ins and determines
  - current position size
  - floating trade profit on the bid-ask spread
  - floating inventory value as the difference between current market value and the bought in value
  - floating P/L as the sum of floating trade profit and floating inventory value

---

<sup>19</sup> Callbacks are a unique feature of the dash framework. Their functionality is explained in section 4.3 but they can be thought of as individual sub-processes.

- De-seasonalized returns: Logic that implements the de-seasoning logic by [Taylor and Xu \(1997\)](#) as described in section 3.2.
- Estimate DCC model parameters and perform step ahead forecast of the Var-Covar Matrix as described in section 3.2.
- Calculate portfolio step-ahead VaR and diversification benefit for each position.
- Enable users to toggle separate tiles of risk related information and historical DCC correlations.

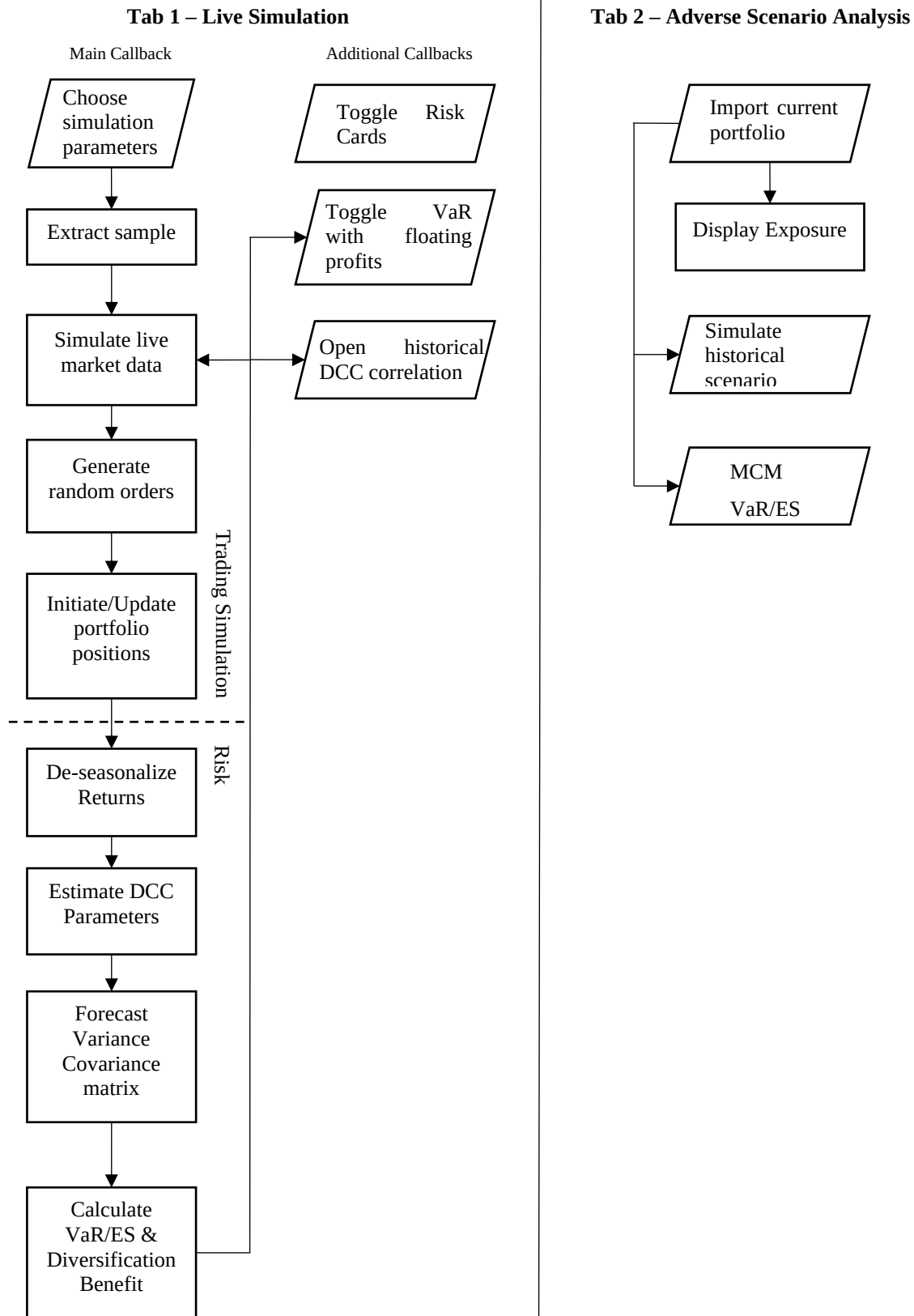
Tab 2:

- Display current exposure of the portfolio including the margin requirement. This gives users the understanding on whether an appreciating or depreciating stock is the tail risk of interest.
- Simulate historic scenario until the end of the trading day with the 5-min average returns of the worst days in the history of the DOW.
- Simulate portfolio values with an MCM to determine the VaR & ES until the end of day.

An additional section of the requirement analysis are the logical requirements of the architecture of the database. The programme runs on a standalone basis without dependencies to other software such as data providers or trading software. The input required is historical data from the pre-sourced sample. Data is only required to be read, written, and stored in a synchronous manner. Therefore, issues of simultaneous access to data are absent. Consequently, handling of an additional database is not necessary. Furthermore, the sample is relatively small.<sup>20</sup> As a result, the data is stored on a series of csv files.

---

<sup>20</sup> 8033KB for the historical price sample.



**Figure 2:** Flowchart of the app's processes.<sup>21</sup>

<sup>21</sup> A granular flowchart of all processes in the programme is found in Appendix B Figure 20.

### 4.3. Design

The programming language of choice was Python. As an interpreter, high-level, object-oriented language, it is suitable for fast application development ([Python Software Foundation 2022](#)). The wide variety of scientific packages and frameworks provide the necessary building blocks to write more complex scripts. According to HackerRank, a technology hiring firm, python lists amongst the top three requested languages in job interviews ([Trikha N.d.](#)). This could improve software maintenance due to familiarity with the language in case of deployment of the programme in a company.

Another reason for Python is its compatibility with the Dash Framework developed by Plotly. Dash. It is an open-source library written on top of Plotly.js<sup>22</sup> and React.js<sup>23</sup> and enables users to natively develop full-stack interactive web applications in either native Python, R or Julia without the additional need for HTML, JavaScript and CSS. Knowledge of these languages helps in case of further customization ([Plotly 2021b](#)). From a company perspective, this enables hired staff familiar with the compatible languages to build proprietary risk management tools without the need for outsourcing as most of the complexity in full-stack development is abstracted away.

Another benefit of Dash is that it capitalises on recent trends towards cloud computing and web-based applications. Dash apps can be deployed as regular flask apps and hosted on virtual private servers. As an application running in the browser, it is cross-platform compatible and mobile ready. In addition, cumbersome software installation rollout on company networks is omittable.

Another benefit of the framework from a risk management perspective is its stateless design. The server is not reliant on states and is self-reliant. In case of a server crash, the server can simply be restarted without dependency on client states. This can reduce outage times.

The Dash framework consists of three core pillars. This is the app.layout, Dash Components and Dash Callbacks. The app.layout structures the position and design of all components in the front-end user interface. The second pillar are the Dash Components themselves. Dash contains a library of components with varying degrees of customization. Dash Callbacks are functions that update Dash components based on one or more input components. A simple example would

---

<sup>22</sup> Plotly.js is a high-level, declarative charting library ([Plotly 2021d](#)).

<sup>23</sup> React.js is a open-source JavaScript library written and maintained by Meta ([2021](#)).

be a date range slider selection updating the date axis of an output graph. Further examples are described in Dash's documentation ([Plotly 2021a](#)).

Langer (2016, p.5-6) argues that the design phase typically does not precede the coding phase entirely in a waterfall type sequence. In reality, it is a trial-and-error process in which both phases occur in tandem. This was also the case in this project. The development process included pencil and paper drawings as well as trial-and-error approaches upon usage of Dash. Screenshots of the final user interface are found in Appendix B Figure 16 onwards. The Callback logic may be extrapolated from the written app code ([Meiners 2022](#)).

#### **4.4. Code**

The choice of python means that certain decisions about programming paradigms are already made. Python follows the 'structured' paradigm. The language is therefore written in a manner that emphasizes clarity and readability of programmes to improve the development process. The use of structured control flow such as conditionals, looping and branching enables better comprehension and reproducibility of larger projects at a future point in time.

The inclusion of functions enables abstracting away complexity and breaking down problems into smaller subsets. This is helpful in the context of this project as it includes a mathematical model that involves multiple steps. Certain variables and functions were required outside of (nested) functions and were required to be declared as global variables. Their implementation requires caution as they have potential to interfere with the global namespace. This can produce bugs that are not easy to trace as the size of the project increases.

The programme is not written from scratch. The main DCC MGARCH logic was sourced from Dr. Christian Contino's github page ([Contino 2021](#)). It was adapted to accommodate the projects requirements such as the forecasting of the var-covar matrix. Other elements include formatting related elements from Dash's documentation. The adapted sections' locations are marked in the app project code ([Meiners 2022](#)).

### **5. Implementation**

Section 4.4 describes the implementation of the app with explanations of the building blocks of the programme with a focus on risk modelling. The process of how the trading simulation operates is not described as its purpose is merely to give the programme a peripheral structure. The sample is also defined in this step.

## 5.1. Sample

### 5.1.1. Prices

The sample consists of the 30 companies in the DOW as of 18.10.2021. This includes 676 trading days that range from 09.12.2019 to 15.10.2021. The range enables app users to choose dates within a range of one-year with sufficient historical data for modelling. The timeframe also matches that of the sources volume data. As volume was sourced at a later stage in the research process, the respective source had a 730-calendar day restriction for historical data which facilitated cropping both the price and volume sample to the same starting date.

The price data follows 5-minute intervals from 09:30 to 16:00 EST. This frequency is chosen in order to avoid microstructure errors in the market (Bollerslev and Andersen 1988). The price data stems from IEX cloud, an API service offered by IEX (IEX Cloud 2021).

The sample was inspected for anomalies and the following data curation was performed. The data consisted of identical prices for 15:55 and 16:30 for every day in the sample. Therefore, the respective returns were 0. To avoid this misestimation of volatility towards the end of the trading day, the 16:30 prices were removed from the sample. There were other shorter occasions of series of identical 5-min prices. These were not removed. The structure of the original programmed script did not allow for days with varying length so days with varying length of data were removed. The model script has since been revised and can accompany 5-min frequency data of different length per historical day. Table 3 provides an overview of the adjustments.

**Table 3:** Price Sample Curation.

| <b>Finding</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Action</b>                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Identical prices on last two trading frequencies                                                                                                                                                                                                                                                                                                                                                                                                                                     | Last traded price removed               |
| Shorter trading days:<br>24.12.2019<br>27.11.2020<br>24.12.2020                                                                                                                                                                                                                                                                                                                                                                                                                      | Day removed                             |
| Occasions of identical prices for periods in the day:<br>03.03-04.03.2020 - 7 5-min frequencies with the same price<br>26-29.06-2020 - 5 5-min frequencies with the same price<br>02.09.2020 - Collection of similar prices towards the end of the trading day<br>26.03.2021 - No trading in 5 securities throughout the day<br><br>Further occasions existed where prices of individual stocks are identical for consecutive 5-min periods. These are not listed individually here. | No Action                               |
| 14.10.2019 - no prices available for Dow (Dow Inc.) security.                                                                                                                                                                                                                                                                                                                                                                                                                        | Completed with price of subsequent day. |

|                                                                  |                                                        |
|------------------------------------------------------------------|--------------------------------------------------------|
| NaN values:<br>83 cells (0.007% of data)<br>Null values:<br>None | Completed with<br>consecutive value of<br>price series |
|------------------------------------------------------------------|--------------------------------------------------------|

### 5.1.2. Volume

Volume data was sourced from the Yahoo Finance API. The highest frequency available was hourly data. Historical data was limited to 730 calendar days as previously mentioned. The earliest possible date with this restriction at the time of sourcing was the 09.12.2019. The sample ranges until the 15.10.2021 to match the price sample. The days of volume and prices must match for the VWAP approach, explained in section 3.2, so the sample was cropped on row index match. 12 datapoints (0.01% of data) had NaN values and 127 (0.12% of data) had null values. These were both completed with consecutive values.

### 5.2. Intraday Seasonality

The new simulated market prices now enable calculating returns required as input for the DCC MGARCH model. The function is constructed to satisfy the de-seasonalizing approach by [Taylor and Xu \(1997\)](#), previously described in section 3.2.

```

1 #function
2 def seasonality(prices):
3     #compute returns
4     rets=prices.pct_change().iloc[1: , :]
5     #1.squared returns
6     squared_rets=(rets**2)
7     #2.mean squared returns per n intraday period
8     for n in range(78):
9         nthret=squared_rets.iloc[n::78, :]
10        if n==0:
11            means=nthret.mean(axis=0).to_numpy()
12        else:
13            means=np.vstack((means,nthret.mean(axis=0).to_numpy()))
14    #3.Divide rets by seasonality term
15    rets_deseason = rets.copy()
16    for n in range(78):
17        rets_deseason.iloc[n::78, :] /= np.sqrt(means[n])
18    #4. Scale rets back to original rets
19    rets_deseason_rescaled=rets.mean()+((rets_deseason-rets_deseason.mean())*(rets.std()/rets_deseason.std()))
20
21    return rets_deseason_rescaled

```

**Figure 3:** De-seasonalizing Function according to [Taylor and Xu \(1997\)](#).

The function in Figure 3 takes prices as inputs formatted as a pandas DataFrame. In the first for loop, squared returns are averaged over the price series for every 5-minute period of a day. The second for loop divides the raw returns by respective seasonality term,  $S_{t,n}$ . See (13).

This step is repeated after each iteration with new simulated market data and the previous 5 days of data. Note how the magnitude of returns in Figure 4, subplot 3 is elevated with the implementation of [Taylor and Xu's \(1997\)](#) method as returns are standardised with mean 0 and standard deviation 1. For estimating risk, this is problematic as returns of greater magnitude

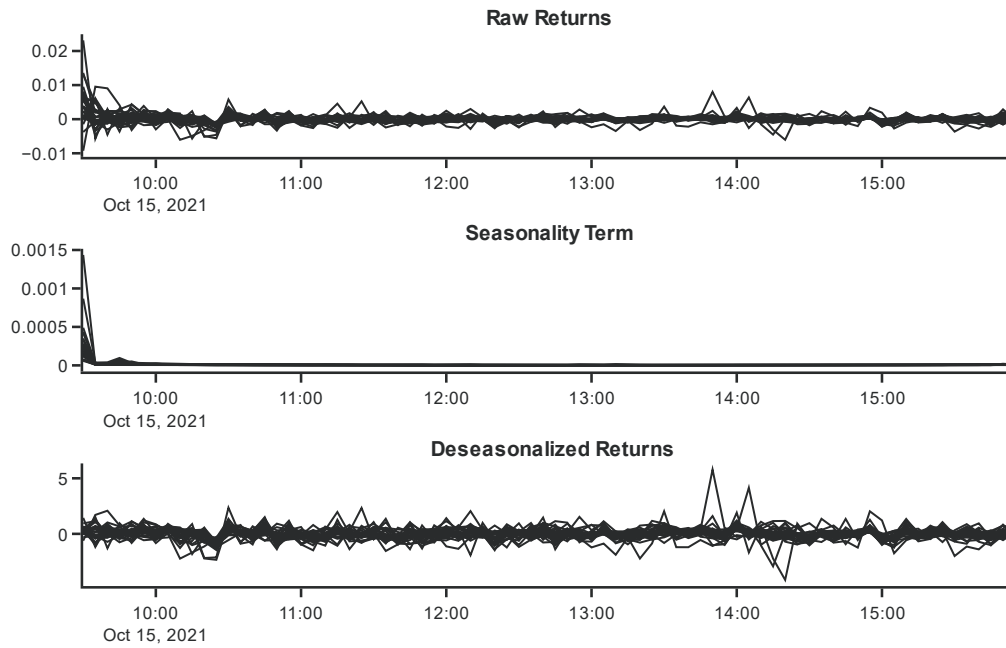


inflate variances and covariances. Therefore, the function in Figure 3 has been extended [line 19] with the formula to rescale the de-seasonalized returns to the magnitude of the original return series:

$$y_i = m_2 + (x_i - m_1) \frac{s_2}{s_1} \quad (28)$$

where  $y_i$  is the new return,  $m_2$  the target mean,  $m_1$  the current mean,  $x_i$  the return to be scaled and  $s_1, s_2$  the standard deviations of the current and the target series respectively.

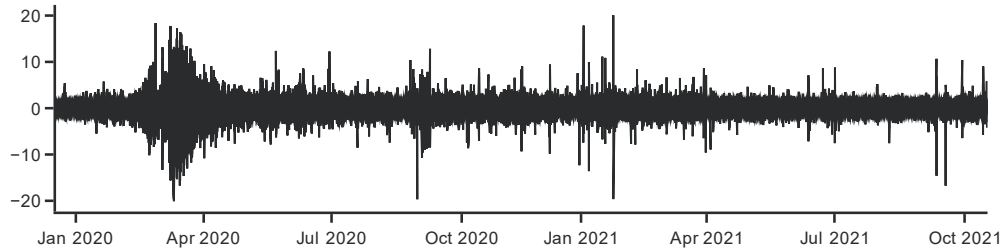
Figure 4 below shows the effect of de-seasonalizing sample returns on the 15<sup>th</sup> of October 2021. Raw returns are elevated during market open while remaining flatter for the rest of the day. This is also reflected in the seasonality term plot. Therefore, the de-seasonalized return series removes the intraday seasonality component.



**Figure 4:** Outcome of de-seasonalizing returns of a day in the sample.

This effect is visible in the ACF plots (see Appendix A Figure 13). All return series exhibit intraday seasonality and Taylor and Xu's (1997) approach removes the seasonality in all cases. It is important to stress that the seasonality term in both Figure 3 and the ACF plots is estimated from the entire sample space as opposed to the selection predefined in the programme's modelling.

While intraday seasonality is removed, the conditional heteroskedastic property of the return series remains as seen in Figure 5. For the modelling process, this means that the model will react to unforeseen changes in volatility and successfully omits potential miscalibrations resulting from intraday seasonality patterns.



**Figure 5:** De-seasonalized return series across sample space.

As previously mentioned, the VWAP approach to de-seasonalizing did not materialize in an effective removal of seasonality. See Appendix A Figure 14. The VWAP approach smoothens the original raw return series. However, high volumes on market open cause higher magnitude of returns. As a result, returns after the VWAP approach remain elevated on market open and remain rather flat for the remainder of the day.

Upon incorporation of the VWAP methodology in the DCC modelling process on the 15h October 2021, the DCC  $\alpha$  (shock) parameter was elevated on market open at a value of around 0.05 indicating a period of market stress, whereas this can be considered a calm day and is merely the increased volatility that is usual for this period of the day.

### 5.3. DCC multivariate GARCH

Once the return series is de-seasonalized, it can safely be used as input for the DCC MGARCH model. Following the logical steps in the code can pose a challenge due to a variety of functions and nested function calls. The following explanation aims at clarifying how the main functions work in the programme. As previously mentioned, the core script was written by Contino (2021), with some modifications to incorporate forecasting, renaming of variables and restructuring to fit into the programme's logical structure.

Figure 6 shows the first step of obtaining both uniformly distributed standardized residuals from the univariate GARCH(1,1) process as well as saving a list of forecasted conditional variances using the `res.forecast` function of the `arch` package [line 14]. The last obtained value is later required for the forecasted variance-covariance matrix. The `for` loop is required to iterate through all securities in the portfolio.

```

1 #initialized empty lists
2 udata_list = []
3 fcast_conditionalvariance_list = []
4
5 #function
6 def run_garch_on_return(rets, udata_list):
7     for x in rets: #forLoop iterates through rets columns
8         am = arch_model(rets[x], dist = 't', p=1, q=1) #garch(1,1) model with t-distribution
9         res = am.fit(disp='off') #regression output of tickers in returns dataframe
10        udata = garch_t_to_u(rets[x], res) #uniformly distributed standardized residuals of return series
11        udata_list.append(udata)
12
13        #forecast variance for t+1 / needed for vector Dt in forecasting DCC function
14        fcastvariance = res.forecast(horizon=1, reindex=False).variance
15        fcastvariance=fcastvariance['h.1'].iloc[0]
16        fcast_conditionalvariance_list.append(fcastvariance)
17
18    return udata_list,fcast_conditionalvariance_list
19
20 #function call
21 udata_list, fcast_conditionalvariance_list = run_garch_on_return(rets.iloc[:, :numvars].dropna(),
22                                                                udata_list)

```

**Figure 6:** Univariate GARCH(1,1) computation according to [Contino \(2021\)](#).

Figure 6 shows the function called in the `run_garch_on_return` function above. SciPy's `t.cdf` function calculates the student-t's cumulative distribution function [line 7].

```

1 def garch_t_to_u(rets, res):
2     mu = res.params['mu'] #mean return
3     nu = res.params['nu'] #kurtosis
4     est_residuals = rets - mu #estimated error term/residuals
5     cond_vol = res.conditional_volatility #conditional volatility
6     std_residuals = est_residuals / np.sqrt(cond_vol) #standardized residuals
7     udata = t.cdf(std_residuals, nu) #uniformly distributed cdf of standardized residuals
8
9     return udata

```

**Figure 7:** Function transforming t-distributed standardized residuals to uniformly distributed residuals according to [Contino \(2021\)](#).

The uniformly distributed standardized residuals stored in the `n` dimensional array, `udata`, can now be used to obtain parameter  $\alpha$  and  $\beta$  as explained in section 3.1.4. Figure 8 shows how this is implemented in the code. As a gaussian copula is implemented according to (10) and (11), SciPy's `norm.ppf` function is required to transform the uniformly distributed standardized residuals to a univariate normal inverse CDF,  $\Phi^{-1}$ . These are stored in the variable `trdata` [line 6]. [Line 10-13] implements the log likelihood estimation from (12). The equivalent of maximising variable `llf` is minimizing `-llf`. In order to optimize for  $\alpha$  and  $\beta$ , bounds and constraints are applied according to the model. [Line 24] implements the constraint that satisfies  $\alpha + \beta < 1$ . The bounds in [line 25] are not a formal requirement but help the optimizer test the parameters on a range that are typically for the model.  $\alpha$  usually ranges from values approaching 0 in a period of calm markets to 0.5 in times of market stress.  $\beta$  usually varies between 0 and 1. The initial guess is defined in [line 19]. While the simulation runs, the optimized values are stored and reused with variable `lasttheta` [line 21] to improve the optimization time.

SciPy's optimization function is called in [line 28]. As constraints and bounds exist, SciPy's default algorithm is SLSQP. It is a sequential least squares programming algorithm using the

Han-Powell quasi-Newton method with a BFGS update (PyOpt 2011; Kraft 1988). The opt\_out variable [line 28] holds the optimized  $\alpha$  and  $\beta$  parameter. The algorithm with input values around the magnitude of 1. Hence, the return series was scaled by factor 1000.

```

1  #function
2  def loglike_norm_dcc_copula(theta, udata):
3      N, T = np.shape(udata)
4      llf = np.zeros((T,1))
5      #std. residuals (marginals of copula) are normally distributed (norm.ppf) for dependency
6      trdata = np.array(norm.ppf(udata).T, ndmin=2)
7
8      Rt, vec1Rt = dcceq(theta,trdata)
9
10     for i in range(0,T):
11         llf[i] = -0.5 * np.log(np.linalg.det(Rt[:, :, i]))
12         llf[i] = llf[i] - 0.5 * np.matmul(np.matmul(trdata[i, :], (np.linalg.inv(Rt[:, :, i]) - np.eye(N))), trdata[i, :].T)
13     llf = np.sum(llf)
14
15     return -llf
16
17 #initialguess
18 if lasttheta==None:
19     initialguess=[0.01,0.95]
20 else:
21     initialguess=lasttheta
22
23 #optimizer constraints and bounds
24 cons = ({'type': 'ineq', 'fun': lambda x: -x[0] -x[1] +1})
25 bnds = ((0.0, 0.5), (0, 0.9997))
26
27 #function call
28 opt_out = minimize(loglike_norm_dcc_copula, np.array(initialguess), args = (udata_list,), bounds=bnds, constraints=cons)

```

**Figure 8:** Log likelihood optimization of Gaussian Copula according to Contino (2021).

Figure 9 shows the functionality of the dcceq function called in the previous step. It is worth noting that initiating empty python lists [line 2-3] for collecting array data in a for loop is faster than populating empty NumPy arrays as NumPy arrays are copied in each iteration of the for loop, demanding more memory. Usually, list comprehension in python is considered slower than NumPy array operations but this depends on the specific use case. The function implements (5), (6), (7) and (8). The script was extended to capture  $\mathbf{Q}_{t-1}$  [line 33-36] needed for forecasting  $\mathbf{H}_t$ .  $\bar{\mathbf{Q}}$  [line 18] has been declared as a global variable as is later required outside of the nested function for forecasting.

```

1 #initialized empty lists
2 lastQt=[]
3 immediatedisturbances_list=[]
4
5 #function
6 def dcceq(theta,trdata):
7     T, N = np.shape(trdata)
8     a, b = theta
9     if min(a,b)<0 or max(a,b)>1 or a+b > .999999:
10         a = .9999 - b
11     Qt = np.zeros((N, N ,T))
12     Qt[:, :,0] = np.cov(trdata.T)
13     Rt = np.zeros((N, N ,T))
14     vec1Rt = np.zeros((T, int(N*(N-1)/2)))
15     Rt[:, :,0] = np.corrcoef(trdata.T)
16
17     for j in range(1,T):
18         global Qbar #use outside nested function for forecast
19         Qbar=Qt[:, :,0]
20         Qt[:, :,j] = Qbar * (1-a-b)
21         immediatedisturbance=np.matmul(trdata[[j-1]].T, trdata[[j-1]])
22
23         Qt[:, :,j] = Qt[:, :,j] + a * immediatedisturbance
24         Qt[:, :,j] = Qt[:, :,j] + b * Qt[:, :,j-1]
25         Rt[:, :,j] = np.divide(Qt[:, :,j],
26                               np.matmul(np.sqrt(np.array(np.diag(Qt[:, :,j])), ndmin=2)).T,
27                               np.sqrt(np.array(np.diag(Qt[:, :,j])), ndmin=2)))
28
29     for j in range(0,T):
30         vec1Rt[j, :] = vec1(Rt[:, :,j].T)
31
32     #capture last optimized qt value of timeseries
33     Qt_end = Qt.flatten()
34     Qt_end=Qt_end[(len(rets)-1)::len(rets)]
35     Qt_end=Qt_end.tolist()
36     lastQt[0:(numvars**2)]=Qt_end
37
38
39     return Rt, vec1Rt

```

**Figure 9:** DCC equation implementation according to [Contino \(2021\)](#).

With the forecasted univariate variance from the GARCH(1,1), the optimized DCC parameters  $\alpha$  and  $\beta$  and the  $\mathbf{Q}_{t-1}$ , the DCC formula can be reimplemented to derive the forecasted variance covariance matrix  $\mathbf{H}_t$  as implemented in Figure 10. The Htforecast [line 22] variable can then be used to derive VaR as shown in section 5.5.

```

1 #Determine Dt for forecasting
2 Dt=np.sqrt(np.diag(np.array(fcast_conditionalvariance_list)))
3
4 #Obtain lastQt
5 listoflists=[]
6 for i in range(0, len(lastQt), numvars):
7     listoflists.append(lastQt[i:i+numvars])
8 lastQt=np.array(listoflists)
9
10 #Set zero vector for error terms
11 fcast_immediatedisturbances=np.zeros((numvars,numvars))
12
13 #DCC's Qt formula
14 Qtforecast=Qbar * (1-a-b) + a * fcast_immediatedisturbances + b * lastQt
15 Qtforecast = np.reshape(Qtforecast, (numvars,numvars))
16
17 #Forecast conditional correlation, Rt
18 Rtforecast = np.divide(Qtforecast , np.matmul(np.sqrt(np.array(np.diag(Qtforecast), ndmin=2)).T,
19                                             np.sqrt(np.array(np.diag(Qtforecast), ndmin=2))))
20
21 #5.Derive forecasted conditional variance-covariance matrix, Ht
22 Htforecast=np.matmul(np.matmul(Dt, Rtforecast),Dt)

```

**Figure 10:** Variance-Covariance Matrix forecast implementation.

#### 5.4. Model Evaluation

As mentioned, univariate model fit has not been implemented within the programme so far. However, a table overview of the univariate GARCH parameters and their respective p-values was included. This enables the user verification on whether the GARCH fit is plausible. By inspecting the alpha and beta coefficients, the user can understand whether the GARCH process in fact uses short term innovations and a long-term flexible lag. Beta coefficients typically take values of around 0.8. In addition, alpha coefficient should not take values of 0 as the modelled volatility would simply be a homoscedastic process. The coinciding p-values give users an understanding on whether the parameters are statistically significant. If strong deviations from typical GARCH models values exist for several univariate fits, it indicates that the resulting multivariate copula and overall DCC MGARCH model fit may be misspecified.

Regarding the multivariate fit, it was not feasible to perform the Kupiec test within the app as it requires roughly 30 minutes to calculate historical scenarios. Therefore, it was performed separately on distinct subsamples. There were chosen based on volatility. As a proxy, the CBOE Volatility Index (VIX), tracking the S&P 500, was chosen to identify suitable periods for testing. This includes both a base case with low levels of volatility and three more scenarios in which the levels of the VIX increase over the course of the backtesting window. The rationale behind this is to test how the model performs in a heteroskedastic environment amid changes in volatility. The scenario's volatility spikes are also visible in Figure 5. An overview of the scenarios is shown in Table 4 below:

**Table 4:** Backtest scenarios.

| Scenario | Time Period   | VIX     |
|----------|---------------|---------|
| Calm     | 08-14.10.2021 | < 20    |
| Vol1     | 23-29-10.2020 | 27 → 40 |
| Vol2     | 25-29.01.2021 | 23 → 37 |
| Vol3     | 07-13.05.2021 | 17 → 27 |

As a further parameter of interest was the choice of data used as an input. First, the length of the time-series used for modelling has impacts on the accuracy of the model, see [Ng and LAM \(2006\)](#). Second, the speed at which parameters are optimized increases exponentially. To initially assess this behaviour, a backtest<sup>24</sup> was performed under modelling of different input lengths. The structure of the output of the backtested scenarios is seen in Appendix A Table 7. The result is shown in Table 5 below:

---

<sup>24</sup> Only evaluating exceedances, not the Kupiec test score.

**Table 5:** Backtested exceedances for  $\text{VaR}_{\alpha=0.05}$  with different input days during the calm market scenario with normally distributed MCM draws.

| Model input days | Average parameter optimization (seconds) | Exceedances | Exceedance threshold at Kupiec alpha 0.05 |
|------------------|------------------------------------------|-------------|-------------------------------------------|
| 5                | 3.780                                    | 22          | 19.45                                     |
| 15               | 12.411                                   | 24          |                                           |
| 25               | 14.519                                   | 23          |                                           |
| 35               | 56.03                                    | 25          |                                           |
| 45               | 86.368                                   | 25          |                                           |

According to Table 5, despite more input data and exponentially longer average optimization time, the resulting exceedances of VaR does not improve. This is despite Ng and LAM's (2006) conclusion that 1000 datapoints are required as a minimum for (daily) GARCH models. As this backtest requires 20 hours to complete, further backtests were only conducted with 5 modelling days.

The subsequent backtests were performed calculating VaR based on the MCM using both normally- and t-distributed random draws. The degrees of freedom were estimated by applying the weighted average of the univariate t-distributed errors of the GARCH(1,1) model. The results are shown in Table 6 and discussed in section 6.1.

### 5.5. Value at Risk

Figure 11 shows the implementation of the 5-min step ahead MCM VaR and ES for tab 1. The user can toggle to include the floating profits from the bid-ask spread in the trading simulation. In case, of a long-short portfolio that has a negative exposure, a margin requirement is added to reflect the additional risk of a margin call [line 11]. The margin requirement can be changed in the parameters before simulation. The default is set to 80%. In the US, the initial Fed requirement on short positions is 50%. Exchanges typically require around 30% (Fidelity 2021). In addition, brokerages may require further margin such that initial margin requirements can amount to 150%. In practice, maintenance margin is required in addition, dependant on the path of the value of the security in question. For simplicity, the initial margin set for the simulation is assumed to be required throughout the lifetime of the short position. The DDOF's of the univariate GARCH fits are used to determine a weighted average DDOF of the portfolio [line 21]. This serves as the basis for the t-distributed random draws of the MCM [line 32].

```

1 def portfolioVaR(df_positions,rets,covMatrix, ddofs_df):
2     mc_num=5000
3     meanrets=(rets.mean(axis=0)).to_numpy()
4     meanrets=np.reshape(meanrets, (len(meanrets),1))
5     covMatrix = np.array(covMatrix)
6     ttoendofday=1 #only step ahead-VaR
7     nstocks=df_positions.loc[:, 'Size'].values
8
9     bool_ind = nstocks < 0
10    #increasing short position by marginrequirement
11    nstocks = np.where(bool_ind ==True, nstocks*(1+marginrequirement), nstocks)
12
13    lastprice=df_positions.loc[:, 'Price'].values
14
15    #determine weighted average of univariate ddofs form univariate garch fit
16    weights_df=(df_positions['Notional'].abs())/(df_positions['Notional'].abs().sum())
17    weights=weights_df.values
18    ddofs=np.delete(ddofs_df.values[0], [0])
19
20    #used subsequently for random draws
21    ddof_weightedavg=np.round(np.sum(np.multiply(weights,ddofs)),3)
22
23    portval=np.sum(np.multiply(lastprice,nstocks))
24    portval=np.repeat(portval, mc_num)
25
26    cholesky = np.linalg.cholesky(covMatrix)
27
28    #Loop not necessary but recovered from app script and not modified much
29    counter=0
30    for i in range(0, ttoendofday):
31        #draw random normal returns
32        draw=np.random.standard_t(ddof_weightedavg, mc_num) #average univariate ddof from portfolio
33        for x in nstocks[:-1]:#Less one as first draw already completed
34            draw=np.vstack((draw, np.random.standard_t(ddof_weightedavg, mc_num)))
35        #correlated random returns
36        corrdraw=np.matmul(cholesky, draw)
37
38        #Add expected returns
39        corrdraw_plusexpret=np.add(meanrets, corrdraw)
40        corrdraw_plusexpret=1+corrdraw_plusexpret #for price multiplication
41
42        #price development of individual stocks
43        if counter==0:
44            stockpricechanges=(corrdraw_plusexpret.T * lastprice).T
45            lastprice=stockpricechanges
46
47        else:
48            stockpricechanges=np.multiply(lastprice,corrdraw_plusexpret)
49            lastprice=stockpricechanges
50
51        stockpricenotionals=(stockpricechanges.T * nstocks).T
52
53        portval=np.vstack((portval,stockpricenotionals.sum(axis=0)))
54
55        counter+=1
56
57    confidencealpha=np.quantile(portval[1], alpha)
58    lastportvalue=portval[0][0]
59
60    #VaR
61    ValueAtRisk=np.round(np.sqrt((confidencealpha-lastportvalue)**2),2)
62
63    #ES
64    belowvar_portvals=portval[1][portval[1]<confidencealpha]
65    ES=np.round(np.sqrt((np.average(belowvar_portvals)-lastportvalue)**2),2)
66
67    floatingprofit=df_positions['P/L'].sum()
68    ValueAtRisk_flprofits=ValueAtRisk-floatingprofit
69    ES_flprofits=ES-floatingprofit
70
71    return ValueAtRisk, ValueAtRisk_flprofits, ES, ES_flprofits
72

```

**Figure 11:** Step-ahead MCM VaR & ES estimation script.

The MCM is implemented according to section 3.4. It is necessary to simulate the price developments of individual stocks first [line 30-51] and aggregate portfolio values afterwards [line 53]. Initially, the code was written to simulate individual assets for time  $t$ , determine the new portfolio value, reallocate assets according to weights, and repeat simulation. This works for long or short only portfolios. However, with a long-short portfolio, the simulated portfolio



values will pass along the portfolio value of 0. If weights are reassigned on each instance, the portfolio values converge to 0 long-term. This creates a path dependence effect. To overcome this, individual asset values are simulated for the whole time series and aggregated on each instance. Confidence intervals for the simulated portfolios are generated with the `np.quantile` function from Numpy and subsequently used to determine VaR & ES.

The function is implemented for the end-of day VaR simulation in the second tab of the app in similar way. The difference being that portfolio values are simulated up to the end of day. As evident from Appendix B Figure 19, an advantage of the MCM is that it provides the risk practitioner with a visual overview of how portfolios and the VaR evolves at the set confidence level throughout the day.

Figure 12 shows the function to calculate the diversification effect of each asset in the portfolio. The forloop iterates such that a parametric VaR is calculated for an individual asset [line 19-26] and a MCM based portfolio VaR is calculated for the rest of the portfolio [line 35] according to Figure 11. The diversification effect is intended to be calculated without expected returns and floating profits. Momentarily however, the VaR function from Figure 11 incorporates expected returns in the random draws [line 39-40]. This can be corrected in future versions of the programme with additional Boolean statements.

```

1 def diveffect(Ht,df_positions,rets):
2
3     df_diversification = pd.DataFrame(columns=['Ticker',
4                                               'Asset_noexpret',
5                                               'Asset_expret',
6                                               'Portfolio_less_asset_noexpret',
7                                               'Portfolio_less_asset_expret',
8                                               'Portfolio_noexpret',
9                                               'Portfolio_expret'
10                                              ])
11     tickervariances=np.diag(Ht)
12     iteration=0
13
14     for ticker in list(df_positions.index.values):
15         rets_iteration=rets.drop(ticker, axis=1)
16         df_positions_iteration=df_positions.drop(index=ticker)
17         ddofs_iteration=ddofs_df.drop(ticker, axis=1)
18
19         if df_positions.loc[ticker]['Notional']>0:
20             ValueatRisk_ticker_noexpret=np.sqrt(
21                 (z_score*df_positions.loc[ticker]['Notional']*np.sqrt(tickervariances[iteration])
22                 )**2)
23         else:
24             ValueatRisk_ticker_noexpret=np.sqrt(
25                 (z_score*df_positions.loc[ticker]['Notional']*(1+marginrequirement)*np.sqrt(tickervariances[iteration])
26                 )**2)
27         iteration+1
28
29         tickerloc=list(df_positions.index.values).index(ticker)
30
31         #strip HT matrix for portfolio Less one
32         Ht_iteration=np.delete(Ht, (iteration-1),axis=0)
33         Ht_iteration=np.delete(Ht_iteration, (iteration-1), axis=1)
34
35         ValueAtRisk_lessticker_nofl,_,_,_=portfolioVaR(
36             df_positions_iteration,rets_iteration,
37             Ht_iteration,
38             ddofs_iteration)
39
40         df_diversification = df_diversification.append({'Ticker':ticker,
41                                                       'Asset_noexpret':ValueatRisk_ticker_noexpret,
42                                                       'Portfolio_less_asset_nofl':ValueAtRisk_lessticker_nofl,
43                                                       'Portfolio_nofl':ValueAtRiskTotal_nofl}, ignore_index=True)
44
45         df_diversification['Div. Benefit_noexpret']=((
46             (
47                 df_diversification['Asset_noexpret']+df_diversification['Portfolio_less_asset_nofl']
48             )-df_diversification['Portfolio_nofl'])/ValueAtRiskTotal_nofl).round(2)
49         df_diversification=df_diversification.set_index('Ticker')
50
51     return df_diversification

```

**Figure 12:** Diversification effect function.

## 5.6. Historical Scenario

The historical scenario is implemented in Tab 2 ‘Adverse Scenario Analysis’, see Appendix B Figure 19. This approach allows for a direct comparison to further evaluate if the MCM based VaR & ES appear plausible and justified compared to the heavy tailed worst historic trading days. This helps the risk practitioner understand how these historic scenarios differ from the MCM simulation with t-distributed random draws.

## 5.7. User Interface

The UI design was built with the use of dash bootstrap<sup>25</sup> cards. Necessary information is grouped within these cards. The user is first able to choose between a set of parameters for the simulation of the programme. The ‘Simulation Status’ card provides feedback on the current

<sup>25</sup> An additional library of components from the open-source Bootstrap library for Dash. Bootstrap is licensed under the MIT License (MIT).

progress of the simulation. Pressing ‘Start’ triggers the main Callback as shown in Figure 2. Once the portfolio has been simulated as described, the modelled VaR & ES, DCC Model parameters, Model Evaluation and the Dynamic Correlation Matrix are displayed. Seeing the Correlation Matrix enable traders to see which assets correlate to determine which securities provide a potential for hedging. The matrix enables users further interactivity. Selecting individual correlations triggers an additional Callback that outputs a graph of the dynamic correlation of the selected securities.

To use output data from the expensive main Callback for other Callbacks, the variables need to be stored. Due to Dash’s stateless design, multiple users will run their own server sessions. Using global variables across processes will not work. The data can be stored with the `ddc.store` component in either the user’s browser session, on a hard drive or server-side memory. In this programme, the former is implemented ([Plotly 2022c](#)).

Appendix B Figure 20 shows the programme’s dash components. The logic of script progresses from left to right. Grey boxes are dash components that contain general component properties like id and name and more specific properties such as conditional formatting for visual elements or the interval frequency for an interval component. Green boxes are the Callbacks (functions). Grey arrows are Callback inputs, blue arrows Callback outputs and dotted lines input of states. States are stored properties or variables that are used within the Callback, but do not trigger the Callback themselves as inputs do.

The programme also incorporated some open source components such as conditional formatting examples ([Plotly 2022a](#)) and a solution for a Start, Stop, Resume logic shared in the Plotly community forum by [User Jwolf \(2022\)](#).

## 6. Evaluation

### 6.1. Risk

To assess whether the programme works with the intent of providing plausible risk forecasts, the outcome of the Kupiec test for multivariate fit is evaluated. The output is shown in Table 6. For the four scenarios on which the model was tested, applying the MCM with t-distributed draws leads to superior outcomes. This choice lowered exceedances in all cases and achieved statistically significant step ahead VaR forecasts at the 95% confidence level in three out of four scenarios. Evaluation of the  $\text{VaR}_{\alpha=0.95}$  at the Kupiec 99% confidence was also performed but failed in all instances.

**Table 6:** Kupiec Backtest Result.

| Backtest length: 5 trading days<br>Observations: <sup>26</sup> 389<br>VaR & Kupiec alpha=0.05<br>Chi squared critical value=3.8415 |          |             |                              |                |
|------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|------------------------------|----------------|
|                                                                                                                                    | Scenario | Exceedances | Test Statistic <sup>27</sup> | 95% Confidence |
| <b>Normal Dist.</b>                                                                                                                | Calm     | 23          | 0.202                        | X              |
|                                                                                                                                    | Vol1     | 42          | 15.681                       |                |
|                                                                                                                                    | Vol2     | 36          | 8.682                        |                |
|                                                                                                                                    | Vol3     | 19          | 0.057                        | X              |
| <b>T-Dist.</b>                                                                                                                     | Calm     | 19          | 0.057                        | X              |
|                                                                                                                                    | Vol1     | 31          | 4.197                        |                |
|                                                                                                                                    | Vol2     | 25          | 0.779                        | X              |
|                                                                                                                                    | Vol3     | 18          | 0.258                        | X              |

Appendix A Figure 15 shows VaR exceedances across time for the backtest. Notable clusters of exceedances are visible towards the end of the Vol1 backtesting scenario. The other scenarios appear to be relatively spread out across the backtesting window.

Overall, the results show promising results that the t-dist. implemented model provides accurate  $\text{VaR}_{\alpha=0.95}$  forecasts in 75% of backtested trading windows. In other words, if a trader faces the choice of using the programme, the ambiguity in assessing risk on the portfolio level on a 5-min intraday basis is reduced. The metrics should however be viewed with caution as they do not work in all instances.

<sup>26</sup> 78 intraday datapoints for 5 days less one for initial return:  $((5*78)-1)$ .

<sup>27</sup>  $2\ln$  of the likelihood ratio in Formula (18).

Further testing could be applied on more market phases with different heteroskedastic properties. Further considerations for improving the modelling process are discussed in section [7.1](#).

The researcher should however aim for a balanced approach as over calibrating the model to yield statistical significance in all past market phase subsamples may lead to overfitting. As a common caveat of backtesting, the two-year sample is unique and will not repeat going forward. The structure of the market and heteroskedastic patterns may change going forward such that a modelled accuracy is not replicable.

Regarding the user experience of programme, the minimalistic interface allows assessing the portfolio and its risk adequately. On the first tab, the position table allows for sorting depending on the field of interest. Key metrics regarding the DCC model approach are also portrayed. The DCC correlation can be inspected visually with a correlation matrix and historic charts. Cards can be opened and collapsed at the user's discretion. On the second tab, users can understand the risk of their position if held until the end of the market day.

## **6.2. Performance**

Assessing the computational performance is also an important part to assessing how suitable the programme is for real life applications. Performance evaluation itself however is a complex topic in the field of computer science. As the thesis has an economic focus, it is explained in short form.

A significant<sup>28</sup> computational burden lies within the optimization of the DCC MGARCH parameters  $\alpha$  and  $\beta$ . The computer with an AMD Ryzen 5 1600 Six-Core Processor with 3.20 GHz and 16.0 GB of RAM requires roughly 2-4 seconds<sup>29</sup> to optimize the parameters. Whether this is fast is a matter of subjective interpretation. For the goal of the programme to supply a trader or risk manager with up-to-date intra-day figures in a timely manner on a minute-by-minute basis, this is sufficient. For a high frequency algorithmic trading bot that requires calculations on a millisecond basis, it is not.

---

<sup>28</sup> See the optimization time in comparison to the overall computing time of the update process in the simulation status card in Appendix [B](#) Figure [16](#). In this case, 37%.

<sup>29</sup> See Table [7](#).

Basic steps to improve the speed of the programme were already taken by [Contino \(2021\)](#) in advance. For instance, the DCC MGARCH related code linked to Scipy's optimizer was already written with Numpy. The Numpy package implements more efficient memory management, integrates faster programming languages: C, C++ and Fortan. Algebraic computations of Numpy Arrays are also processed in parallel as opposed to regular Python lists that are calculated on a consecutive basis due to the restrictions of Python's Global Interpreter Lock (GIL) ([GeeksforGeeks 2022](#)).

Therefore, most other computational tasks were also performed using the Numpy package. Furthermore, the last optimized values are stored for the next iteration as the optimum parameters are not expected to change significantly for every new 5-minute period in the time series, further increasing the performance on the second optimization onwards. See Figure 8. However, some of the code structure was also implemented with the goal of readability versus optimum performance. For instance, the risk related functions are placed within the Dash Callback where the computation occurs to better understand the logical process of the app. Therefore, the functions are redefined upon every iteration of the programme.

[Morier \(2020, p. 105\)](#) note that the best computational speeds are achieved by directly writing the script in a low-level language such as C or Fortan. However, the Dash framework does not currently support these languages. The author further recommends implementing the Numba package in Python which compiles python code on-the-fly. Choosing Numba would require some restructuring of the code as the currently implemented pandas package is not compatible ([Anaconda, Inc. 2022](#)).

The comments on performance are based on experience with a sample of 30 stocks, how performance differs in cases with several 100 or 1000 stocks has not yet been investigated.

### **6.3. Dash**

From a development perspective, the overall normative impression was that the Dash framework is a powerful tool for creating fully functional applications. The logic behind the structure of the framework accomplishes two things. On the one hand, it is easy for beginners to understand and start creating simple applications. On the other hand, it provides enough creative room to implement more complex interactivity and interdependencies with the possibility to incorporate custom HTML, JavaScript, and CSS. The simplicity allows creating a risk monitoring tool of this thesis in just over 2000 lines of code.

For individual developers or organisations, the fact that HTML, JavaScript, and CSS is abstracted away into pure python means that risk practitioners that are familiar with python as a popular data science language, are now able to build GUI based applications without the requirement of a full stack development team. This could cut cost and reduce the risk of misunderstandings between risk practitioners and developers if the latter is required to a lesser extent or even omitted.

From a deployment perspective, the stateless design enables easy scalability for bigger organisations. The fact that Dash applications run in the browser makes deployment on networks easier as server access is the only requirement for users. Additional installations of programmes are not needed so that roll out and maintenance of the app become easier.

## **7. Limitations**

This section discusses limitation of the modelling approach as well as the technical implementation via the Dash Framework.

### **7.1. Modelling**

This section discusses shortcomings of the standard GARCH model, problems related to higher-dimensional modelling and issues related to the use of high frequency intraday data.

#### **7.1.1. GARCH**

Since the introduction of the GARCH model by [Bollerslev \(1986\)](#), a family of GARCH models have been developed in order to address shortcomings of the original model. It is beyond the scope of this paper to discuss all shortcomings and potential superior models in detail; however, three limitations of the original model are discussed below.

One limitation of the GARCH model is the symmetry in the error term. Therefore, positive returns have the same symmetrical impact on volatility as negative returns. A stylized fact in financial markets however is that negative shocks in returns impact volatility more than positive ones. One of the possible economic rationales<sup>30</sup> behind this phenomenon is the leverage effect. As falling security prices reduce a company's equity/debt ratio, they are perceived as riskier ([Bollerslev 2008](#)). [Nelson \(1991\)](#) introduced the exponential (E-) GARCH model to allow for an asymmetric error term. In the context of this thesis, this means that current VaR estimates may be unnecessarily high in upward trending markets with the standard GARCH model. As a

---

<sup>30</sup> [Wang and Yang \(2017\)](#) note that although GARCH related literature is vast, research on economic origins remains scarce. However, the authors provide a good literature review of economic origins.

result, the risk practitioner could consider combining the DCC MGARCH model with a univariate E-GARCH model as opposed to the standard GARCH model.

Other models also address volatility asymmetry like the Glosten-Jagannathan-Runkle (GJR-) GARCH model ([Glosten et al. 1993](#)). The GJR GARCH model has been extended by ([Wang and Yang 2017](#)) to addresses a further limitation. In addition to volatility asymmetry, persistence in volatility differs based on whether positive or negative shocks occur. The lag-term in the standard GARCH model fails to reflect this stylized fact. They evaluate their extended GJR GARCH model on the S&P 500 ETF and S&P 100 index stocks and conclude improvements in volatility forecasting accuracy.

[Andersen et al. \(2003\)](#) note that GARCH models were initially developed for the use on daily data and are unable to safely accommodate information contained in intraday data. The specifications of the model result in overshooting of risk figures that are too slow mean revert, failing to capture longer inter-daily volatility. Therefore, they are impractical for market environments marked by fast changes in volatility. The authors postulate a realized volatility (RV) model using the sum of squared returns. This quadratic variation serves as an unbiased and efficient estimator for return volatility [Andersen et al. \(2003\)](#). They find superior forecasting performance compared to both conventional<sup>31</sup> daily volatility forecasts and more sophisticated intraday models. Further models have since been developed, one of which is [Hansen et al.'s \(2012\)](#) Realized GARCH framework. It follows a joint modelling approach implementing both an RV- and GARCH component. The authors also find significant improvements in empirical fit on a sample of DOW index stocks compared to standard GARCH.

### **7.1.2. Dimensionality**

RV GARCH addresses a second problem, regarding higher-dimensional environments. When the GARCH or related frameworks are applied to a multivariate setting, the computational expense increases exponentially, a term dubbed as ‘dimensionality curse’. It has led practitioners to shy away from incorporating such models. As RV captures intraday return data without the explicit necessity for modelling, the complexity is reduced and is thus more appropriate in a practical setting ([Andersen et al. 2003](#)). However, this must be viewed in the context of time as almost 20 years have passed since the publication and computing power has risen exponentially since. Nevertheless, although the programme currently performs adequately

---

<sup>31</sup> Standard GARCH and ‘RiskMetrics’ exponential smoothing.



well, the sample only included 30 stocks and it has not been studied up to what extent this applies to significantly larger sample sizes.

Higher dimensions introduce further caveats, one of which may be an introduced bias in the standard estimator. Estimation methods of most models to determine the unconditional variance-covariance matrix includes estimation of  $L^2$  parameters using LT data points, where T is the length of the time-series and L the number of securities. Estimates become biased if T is not substantially larger than L due to a contaminated likelihood function (Pakel et al. 2021).<sup>32</sup> This does not pose a challenge for a sample of 30 securities like the sample size in this research, but requires consideration for significantly larger size of L. Pakel et al. (2021) propose a novel estimation technique addressing this issue and could be considered in the design of the programme.

### 7.1.3. Model Evaluation

The backtest was not performed on the period in the sample with the most extreme volatility shock, the 2020 stock market crash. This is because the backtest produced faulty output for this time window. For most iterations, the Scipy's SLSQP algorithm could not perform due to either one of the following error messages:

- Positive directional derivative for linesearch.
- Iteration limit reached.
- Inequality constraints incompatible.

A possible reason may be that the current script is written with the constraint that it performs on return series of 78 days in length for each day. The 2020 stock market crash experiences circuit breakers. Therefore, prices remain steady in the dataset, causing flat returns that imply low volatility amid a high volatility environment. Perhaps these drastic differences in magnitude of returns are the reason the optimizer struggles to perform.

Regarding backtesting, the formal Christofferson test could also be implemented to test whether the assumption of the IID Bernoulli process of the Kupiec Test is valid. Furthermore, the backtest could be tested in more scenarios to get a more definitive answer on whether the application of the normally distributed- or the t-distributed MCM performs better.

---

<sup>32</sup> Please see source for the full mathematical proof.

In addition, the backtest was only performed on one specific calibration of the DCC MGARCH model. (1) Univariate standard GARCH(1,1) with t-distributed errors with a (2) Gaussian copula for multivariate dependence. (3) Only the DCC MGARCH model applied although other multivariate GARCH models exist. (4) Applied only with the MCM VaR with (5) one specific candidate portfolio (6) on subsamples of a specific 2-year sample with (7) one method of de-seasonalizing returns. Backtesting could thus be repeated with other such specifications.

#### **7.1.4. DCC multivariate GARCH**

One of the drawbacks of the DCC model is that parameters  $\alpha$  and  $\beta$  themselves are scalars. As a result, all conditional correlations behave with the same linear dynamics. It is prerequisite of the model to ensure that vector  $\mathbf{R}_t$  is positive definite. The larger the variable size,  $N$ , the more restrictive the common dynamics become (Bauwens et al. 2006). Billio et al. (2003) introduce a block-diagonal structure for the specification of the  $\mathbf{Q}_t$  vector that allows for variability in the dynamics by grouping variables into blocks. Pelletier (2006) allow for variability in dynamics. Not over the variables themselves, but across time by incorporating a regime switching model that allows for distinct dynamics within each regime.

#### **7.1.5. Seasonality**

The DCC MGARCH parameter inputs (stock returns) currently treat overnight and intraday returns the same despite potential differences in statistical properties. Linton and Wu (2016) stress the difference of distributions between overnight and intraday returns. The former exhibits fatter tails as different quantities of new information are priced into stocks. As a result, the authors develop a model that captures overnight-, intraday- and longer run shocks separately. The authors model shocks via implementation of a dynamic conditional score (DCS) model, that relates news impact curves to the residuals of a t-distribution with unknown degrees of freedom. The authors find an improvement in intraday volatility forecasts.

In the context of VaR forecasting, Fuertes and Olmo (2012) studied the impact of different VaR modelling techniques in the context of intraday returns. They found that including overnight returns in the GARCH modelling process fails to improve daily VaR forecasts. One approach could therefore be to omit them from the timeseries. However, it is important to note that their analysis was not based on the DCC MGARCH model, and their analysis was conducted on daily VaR figures as opposed to an intraday step ahead VaR so their finding might not hold in this scenario.

Another aspect of seasonality are varying degrees of intraday seasonality across time. Research has shown that autocorrelation differs on Mondays and the other trading days of the week due to different levels of price discovery. There are also monthly effects, such as earnings versus non earnings months (Dong et al. 2017). In an attempt to refine the approach, the seasonality term by Taylor and Xu (1997) approach could be further granulated to incorporate different return averages for Monday- versus non-Monday and earnings- versus non-earnings intraday periods.

Voges et al. (2019) study seasonality in the DOW on both the index level and its individual constituents and find differences in memory effects for both realized volatility and trading volume. Deterministic models work well on the index level but not on an individual stock basis. The authors argue that addressing seasonality of individual stocks therefore requires another approach.

Allez and Bouchaud (2011) discover further stylized facts in intraday returns. Average correlations increase as the day progresses, such that early trading hours are marked by idiosyncratic or sector risk while the market factor gains importance for the remainder of the day. Furthermore, kurtosis is low on market open amid volatility peaking. Kurtosis grows as the day progresses. Therefore, volatility surprises become more prevalent in the remainder of the day. These characteristics could also be controlled for by implementing market and industry factors as well as kurtosis prior to the DCC MGARCH modelling process.

#### **7.1.6. Value At Risk**

The current VaR is derived from a gaussian copula whose properties underestimates joint tail dependence in financial markets. Therefore, it is better to resort to more leptokurtic distribution that has superior performance in estimating the low probability regions of the distribution that VaR measures (Angelidis et al. 2004). Using a non-Gaussian copula such as a t-copula could provide better estimate. Modelling fat tails directly with the incorporation of Extreme Value Theory based distributions has also shown superior risk estimations (Karmakar and Paul 2019). The same applies for the choice of distribution for the random simulated portfolios in the MCM.

Limitations of VaR should also be respected and understood by the risk practitioner. Notable limitations are that VaR is (1) not sub-additive. Consequently, the condition  $VaR(V1 + V2) \leq VaR(V1) + VaR(V2)$  is not always satisfied. (2) Past distributions are not indicative of future distributions. (3) Extreme historical observations are rare and hence difficult

to model. (4) Introduced estimation bias<sup>33</sup> if dimensions of the variance-covariance matrix exceeds historical observations ([Krause 2003](#)). To address issue (1), the Expected Shortfall was also included as a complementary risk figure in the app.

## 7.2. Dash

A limitation of Dash is that outputs of Callbacks cannot be used for inputs to invoke a repeated Callback. This would have enabled that once risk figures are calculated in the main Callback, the Callback simply repeats to update the model on a continuous basis.

As this was not possible, a separate interval component is implemented to invoke the Callback every 20 seconds. The first drawback is that if the Callback is faster than the interval component, updating risk figures is unnecessarily prolonged. Second, if the Callback is slower than the interval component, the Callback aborts and certain calculated variables are reused, invoking a bug in the code and failure to calculate new risk figures. The researcher attempted to overcome this problem by implementing a second control type Callback so that the Callback is only calculated when the last Callback has updated the model parameters based on a suggestion by [User jcthomas \(2022\)](#). The approach failed to materialize up to this point.

Furthermore, the use of `dcc.store`<sup>34</sup> components became cumbersome with increased complexity of the app. It becomes challenging to manage more `dcc.store` components as each require additional documentation within Callbacks. Furthermore, the data requires conversion to the JSON format for in-browser storage. Towards the end of the research process, it was found that a solution by a community developer exists that involves storing output components server-side ([User thedirtyfew 2021](#)). Reconsidering this capability would have led to an easier development process.

---

<sup>33</sup> Also mentioned in section [7.1.2](#).

<sup>34</sup> Please note that ‘dcc’ refers to Dash Core Components whereas ‘DCC’ refers to Dynamic Conditional Correlation throughout the thesis.

## 8. Further research

Ideas for further research were derived from the limitations identified in section 7. The programme could be tested for performance and model accuracy with more variables, perhaps 1000 securities, to determine whether the programme's design along with the DCC MGARCH model choice makes for an appropriate real-world implementation. Furthermore, other securities, perhaps from other markets, could also be incorporated to control for potential variability across markets.

Linked to the previous aim of improving performance and accuracy in forecasts, the programme could be extended to incorporate an RV- DCC MGARCH model to compare whether both performance and forecasting ability improves.

In addition, different approaches to modelling dependence within the DCC MGARCH model could be tested and compared. For instance, other copulas such as the Student-t, Skewed student-t or Gumbel could be incorporated.

Concerning the use of high frequency intraday data, different data intervals could also be tested to compare how the model performs on lower or higher frequencies of intraday data compared to 5-min intervals. Furthermore, different methods of de-seasonalizing data could be considered such as the use of FFF or P-GARCH models.

Although the approach of de-seasonalizing data using VWAP in isolation did not result in the desired outcome, it could be incorporated as an additional factor rather than in isolation. Other intra-day variables like the bid-ask spread or volume on first limits of the orderbook could be included as further factors. See [Bińkowski and Lehalle \(2018\)](#). Given that volume data was already sourced for calculating VWAP, it could also be used to model heteroskedasticity in volatility directly. [Wagner and Marsh \(2005\)](#) have included surprise volume (unexpected above-average trading activity) to model persistence in volatility.

From a risk perspective, further analysis could be done to explicitly backtest and compare MCM VaR values and exceedances modelled with both standard variance-covariance matrix and the DCC MGARCH modelled variance-covariance matrix. The MCM based VaR could also be compared with non-parametric VaR calculations that use quantiles, extreme returns or the pareto distribution as implemented by [Giot \(2000\)](#). From a computational perspective, the implementation of multiprocessing to distribute the expensive minimization algorithm across multiple threads as opposed to the default queued executing within python could be considered.

## 9. Conclusion

The thesis evaluates a software solution to monitor risk of stocks on a real-time basis. The risk metric of choice is an MCM VaR using t-distributed random portfolio value simulations. The step-ahead variance-covariance matrix is forecasted via the DCC MGARCH model by [Engle \(2002\)](#). Multivariate joint dependence is modelled with a gaussian copula. [Contino \(2021\)](#) provides the foundation for the programmatic implementation of the model. The application is written with the Dash Framework by [Plotly \(2021c\)](#). Features include the display of a step ahead VaR and Expected Shortfall, both with and without floating profits. These are derived from a trading simulation on a historical sample of DOW's index constituents. In addition, users can simulate portfolios under adverse scenarios in a second tab.

Determining intraday risk adds further complexity in accounting for intraday seasonality. It is removed by incorporating [Taylor and Xu's \(1997\)](#) seasonality term. An attempt to include the VWAP did not yield desired outcomes. The model is evaluated with a Kupiec test that includes back testing the VaR on historic subsamples. These included four 5-day periods, three of which the sample's volatility experienced strong heteroskedastic properties.

The Kupiec test results in [Table 6](#) show that the model's forecasted VaR is statistically significant in three out of four subsample periods.

Writing code with Dash is accessible. The Framework's pure python approach reduces complexity that comes with frontend web development. This enables faster development time and bears a cost saving potential.

Limitations of the research include that the DCC MGARCH model has been calibrated in one specific procedure. Evaluating different univariate models such as RV- or GJR- GARCH and experimenting with further copulas is of future interest. Considering model evaluation, univariate fit analysis such as QQ-plots could still be incorporated into the programme. Regarding multivariate fit, the Kupiec test could be tested on further subsample periods. The Christofferson may also be included as formal approach to evaluating independence between occurrences of VaR exceedances. Although the application currently performs as intended on the tested computer, the sample currently is restricted to 30 securities. Further performance testing is required on more realistic scenarios of 100 to 1000 stocks to evaluate to what extent the 'curse of dimensionality' affects model computation time.

## Publication bibliography

- Allez, R.; Bouchaud, J. P. (2011): Individual and collective stock dynamics: intra-day seasonalities. In *New J. Phys.* 13 (2), p. 25010. DOI: 10.1088/1367-2630/13/2/025010.
- Almeida, D.; Hotta, L. K.; Ruiz, E. (2018): MGARCH models: Trade-off between feasibility and flexibility. In *International Journal of Forecasting* 34 (1), pp. 45–63. DOI: 10.1016/j.ijforecast.2017.08.003.
- Anaconda, Inc. (2022): A ~5 minute guide to Numba. Available online at <https://numba.pydata.org/numba-doc/latest/user/5minguide.html>, checked on 2/5/2022.
- Andersen, T. G.; Bollerslev, T.; Diebold, F. X.; Labys, P. (2003): Modeling and Forecasting Realized Volatility. In *Econometrica* 71 (2), pp. 579–629.
- Angelidis, T.; Benos, A.; Degiannakis, S. (2004): The use of GARCH models in VaR estimation. In *Statistical Methodology* 1 (1-2), pp. 105–128. DOI: 10.1016/j.stamet.2004.08.004.
- Bauwens, L.; Laurent, S.; Rombouts, J. V. K. (2006): Multivariate GARCH models: a survey. In *J. Appl. Econ.* 21 (1), pp. 79–109. DOI: 10.1002/jae.842.
- Bayer, T.; Mai, C.; Speicher-Utsch, S.; Göggelmann, U. (Eds.) (2010): Das Geheimnis des großen Börsencrashes. Financial Times Deutschland. Available online at <https://web.archive.org/web/20100512164019/http://www.ftd.de/finanzen/maerkte/marktberic.hte/1000-punkte-rutsch-das-geheimnis-des-grossen-boersencrashes/50113071.html>, updated on 5/11/2010, checked on 1/30/2022.
- Będowska-Sójka, B. (2015): Daily VaR forecasts with realized volatility and GARCH models. In *Argumenta Oeconomica* 1 (34), pp. 157–173. DOI: 10.15611/aoe.2015.1.06.
- Bera, A. K.; Higgins, M. L. (1993): ARCH Models: Properties, Estimation and Testing. In *Journal of Economic Surveys* 7 (4), pp. 307–366.
- Billio, M.; Caporin, M.; Gobbo, Michele (2003): Block Dynamic Conditional Correlation Multivariate GARCH models. Available online at <https://www.greta.it/old/wp/03.03.PDF>, checked on 1/14/2022.
- Bińkowski, M.; Lehalle, C. A. (2018): Endogeneous Dynamics of Intraday Liquidity. Available online at <https://arxiv.org/pdf/1811.03766>.

- Bollerslev, T. (1986): Generalized autoregressive conditional heteroskedasticity. In *Journal of Econometrics* 31 (3), pp. 307–327. DOI: 10.1016/0304-4076(86)90063-1.
- Bollerslev, T. (2008): Glossary to ARCH (GARCH). In *CREATES Research Paper*, pp. 2008–2049.
- Bollerslev, T.; Andersen, T. G. (1988): Answering the Skeptics: Yes, Standard Volatility Models do Provide Accurate Forecasts. In *International Economic Review* 39 (4), pp. 885–905. DOI: 10.2307/2527343.
- Bollerslev, T.; Andersen, T. G. (1997): Intraday periodicity and volatility persistence in financial markets. In *Journal of Empirical Finance* 4 (2-3), pp. 115–158. DOI: 10.1016/S0927-5398(97)00004-2.
- Bollerslev, T.; Ghysels, E. (1996): Periodic Autoregressive Conditional Heteroscedasticity. In *Journal of Business & Economic Statistics* 14 (2), pp. 139–151. DOI: 10.1080/07350015.1996.10524640.
- Boudt, K.; Croux, C.; Laurent, S. (2008): Robust Estimation of Intra-week Periodicity in Volatility and Jump Detection. In *SSRN Journal*. DOI: 10.2139/ssrn.1297371.
- Campbell, J. Y.; Lo, A. W.; MacKinlay, A. C. (2012): The econometrics of financial markets. [ebook]. Princeton, NJ: Princeton University Press. Available online at <https://ebookcentral.proquest.com/lib/gbv/detail.action?docID=5515221>.
- Chua, D. B.; Kritzman, M.; Page, S. (2009): The Myth of Diversification. In *JPM* 36 (1), pp. 26–35. DOI: 10.3905/JPM.2009.36.1.026.
- Contino, C. (2016): A Bayesian Approach to Risk Management in a World of High-Frequency Data. Doctoral Thesis. The University of Sydney, Sydney.
- Contino, C. (2021): DCC Model [Source Code]. Available online at <https://github.com/tinoproductions/DirtyQuant/blob/master/DCC%20Model.ipynb>, updated on 11/30/2021, checked on 12/26/2021.
- Diebold, F. X.; Lopez, J. A. (1995): Modeling Volatility Dynamics. In K. D. Hoover (Ed.): *Macroeconometrics*. Dordrecht: Springer Netherlands, pp. 427–472.
- Dong, X.; Feng, S.; Ling, L.; Song, P. (2017): Dynamic autocorrelation of intraday stock returns. In *Finance Research Letters* 20, pp. 274–280. DOI: 10.1016/j.frl.2016.10.008.



Engle, R. (2002): Dynamic Conditional Correlation: A Simple Class of Multivariate Generalized Autoregressive Conditional Heteroskedasticity Models. In *Journal of Business & Economic Statistics* 20 (3), 339-350.

Engle, R. F. (1982): Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. In *Econometrica* 50 (4), pp. 987–1002. DOI: 10.2307/1912773.

Engle, R. F. (2001): GARCH 101: An Introduction to the Use of Arch/Garch Models in Applied Econometrics. NYU Working Paper No. FIN-01-030. Available online at <https://web-static.stern.nyu.edu/rengle/GARCH101.PDF>, checked on 1/8/2022.

Everything Explained Today (2021): List of largest daily changes in the Dow Jones Industrial Average explained. Available online at [https://everything.explained.today/List\\_of\\_largest\\_daily\\_changes\\_in\\_the\\_Dow\\_Jones\\_Industrial\\_Average/](https://everything.explained.today/List_of_largest_daily_changes_in_the_Dow_Jones_Industrial_Average/), checked on 1/17/2022.

Fidelity (2021): Trading FAQs: Margin. Available online at <https://www.fidelity.com/trading/faqs-margin#>, checked on 1/17/2022.

Fuertes, A.; Olmo, J. (2012): The Role of High-Frequency Prices, Long Memory and Jumps for Value-at-Risk Prediction. In *SSRN Journal*. DOI: 10.2139/ssrn.2070018.

GeeksforGeeks (2022): Why is Numpy faster in Python? Available online at <https://www.geeksforgeeks.org/why-numpy-is-faster-in-python/>, updated on 8/13/2021, checked on 1/28/2022.

Giot, P. (2000): Intraday Value-At-Risk. Available online at <http://webdoc.sub.gwdg.de/ebook/serien/e/CORE/dp2000-45.pdf>, checked on 1/19/2022.

Glosten, L. R.; Jagannathan, R.; Runkle, D. E. (1993): On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks. In *The Journal of Finance* 48 (5), p. 1779. DOI: 10.2307/2329067.

Hansen, P. R.; Huang, Z.; Shek, H. H. (2012): Realized GARCH: a joint model for returns and realized measures of volatility. In *J. Appl. Econ.* 27 (6), pp. 877–906. DOI: 10.1002/jae.1234.

Hansen, P. R.; Lunde, A. (2005): A forecast comparison of volatility models: does anything beat a GARCH(1,1)? In *J. Appl. Econ.* 20 (7), pp. 873–889. DOI: 10.1002/jae.800.

Harris, L. (1986): A transaction data study of weekly and intradaily patterns in stock returns. In *Journal of Financial Economics* 16 (1), 99-117. DOI: 10.1016/0304-405X(86)90044-9.

Holton, G. A. (2022): Value-At-Risk. [ebook]. Available online at <https://www.value-at-risk.net/backtesting-coverage-tests/>, checked on 1/26/2022.

Humboldt University Berlin, School of Business and Economics (n.D.): Multivariate GARCH models. Available online at [http://sfb649.wiwi.hu-berlin.de/fedc\\_homepage/xplore/tutorials/sfehtmlnode68.html](http://sfb649.wiwi.hu-berlin.de/fedc_homepage/xplore/tutorials/sfehtmlnode68.html), checked on 1/9/2022.

IEX Cloud (2021). Available online at <https://www.iexcloud.io/>, checked on 12/26/2021.

Karmakar, M.; Paul, S. (2019): Intraday portfolio risk management using VaR and CVaR: A CGARCH-EVT-Copula approach. In *International Journal of Forecasting* 35 (2), 699-709. DOI: 10.1016/j.ijforecast.2018.01.010.

Kraft, D. (1988): A software package for sequential quadratic programming. Deutsche Forschungs- und Versuchsanstalt für Luft- und Raumfahrt (Report DFVLR-FR 88-28).

Krause, A. (2003): Exploring the Limitations of Value at Risk: How Good Is It in Practice? In *The Journal of Risk Finance* 4 (2), pp. 19-28. DOI: 10.1108/eb022958.

Langer, M. L. (2016): Guide to Software Development. Designing and Managing the Life Cycle. 2nd edition: Springer-Verlag London.

Linton, O. B.; Wu, J. (2016): A Coupled Component GARCH Model for Intraday and Overnight Volatility. In *SSRN Journal*. DOI: 10.2139/ssrn.2874631.

Martens, M.; Chang, Y. C.; Taylor, S. J. (2002): A Comparison of Seasonal Adjustment Methods When Forecasting Intraday Volatility. In *J Financial Res* 25 (2), pp. 283-299. DOI: 10.1111/1475-6803.t01-1-00009.

Meiners, T. (2022): DCCMGARCH-DASH-APP [Source Code]. Available online at <https://github.com/meinerst/DCCMGARCH-DASH-APP/blob/main/App.ipynb>, updated on 2/11/2022, checked on 2/11/2022.

Meta (2021): React. Available online at <https://reactjs.org/>, checked on 12/24/2021.

Morier, B. (2020): Modelling High Frequency Intraday Discrete Returns. PhD Thesis. Escola de Economia de São Paulo da Fundação Getulio Vargas, Sao Paulo. Available online at [https://bibliotecadigital.fgv.br/dspace/bitstream/handle/10438/29115/\\_Tese\\_PhD\\_FGV%20Bruno%20Morier%20Final.pdf](https://bibliotecadigital.fgv.br/dspace/bitstream/handle/10438/29115/_Tese_PhD_FGV%20Bruno%20Morier%20Final.pdf), checked on 2/5/2022.

Morimoto, T.; Kawasaki, Y. (2008): Empirical Comparison of Multivariate GARCH Models for Estimation of Intraday Value at Risk. In *SSRN Journal*. DOI: 10.2139/ssrn.1090807.

Nelson, D. B. (1991): Conditional Heteroskedasticity in Asset Returns: A New Approach. In *Econometrica* 59 (2), p. 347. DOI: 10.2307/2938260.

Ng, H. S.; LAM, K. P. (2006): How does Sample Size Affect GARCH Models? In : Proceedings of the 9th Joint International Conference on Information Sciences (JCIS-06). 9th Joint International Conference on Information Sciences (JCIS-06). not available, 1/1/1970 - 1/1/1970: Atlantis PressParis, France (Advances in Intelligent Systems Research).

Orskaug, E. (2009): Multivariate DCC-GARCH Model. -with various error distributions. Master Thesis. Norwegian University of Science and Technology, Norway. Available online at [https://ntnuopen.ntnu.no/ntnu-xmlui/bitstream/handle/11250/259296/724505\\_FULLTEXT01.pdf?sequence=2&isAllowed=y](https://ntnuopen.ntnu.no/ntnu-xmlui/bitstream/handle/11250/259296/724505_FULLTEXT01.pdf?sequence=2&isAllowed=y), checked on 1/26/2022.

Packham, N. (2021): Lecture 6: Computational Finance and FinTech Probabilities and Monte Carlo simulation. lecture notes. Berlin School of Economics and Law. Berlin (LV-Nr. 425093).

Pakel, C.; Shephard, N.; Sheppard, K.; Engle, R. F. (2021): Fitting Vast Dimensional Time-Varying Covariance Models. In *Journal of Business & Economic Statistics* 39 (3), pp. 652–668. DOI: 10.1080/07350015.2020.1713795.

Pelletier, D. (2006): Regime switching for dynamic correlations. In *Journal of Econometrics* 131 (1-2), pp. 445–473. DOI: 10.1016/j.jeconom.2005.01.013.

Plotly (2021a): Basic Dash Callbacks. Available online at <https://dash.plotly.com/basic-callbacks>, checked on 12/24/2021.

Plotly (2021b): Dash Open Source. Available online at <https://plotly.com/dash/open-source/>, checked on 12/24/2021.

Plotly (2021c): Introduction to Dash. Available online at <https://dash.plotly.com/introduction>, checked on 10/20/2021.

Plotly (2021d): Plotly JavaScript Open Source Graphing Library. Available online at <https://plotly.com/javascript/>, checked on 12/24/2021.

Plotly (2022a): Conditional Formatting. Available online at <https://dash.plotly.com/datatable/conditional-formatting>, checked on 1/19/2022.

- Plotly (2022b): Dash Enterprise App Gallery. Available online at [https://dash.gallery/Portal/?search=\[Financial\]](https://dash.gallery/Portal/?search=[Financial]), checked on 12/24/2021.
- Plotly (2022c): Sharing Data Between Callbacks. Available online at <https://dash.plotly.com/sharing-data-between-callbacks>, checked on 1/19/2022.
- PyOpt (2011): SLSQP - Sequential Least Squares Programming. Available online at <http://www.pyopt.org/reference/optimizers.slsqp.html>, updated on Feb. 2011, checked on 1/12/2022.
- Python Software Foundation (2022): What is Python? Executive Summary. Available online at <https://www.python.org/doc/essays/blurb/>, checked on 12/24/2021.
- Sheppard, K. (2021): bashtage/arch: Release 5.1.0: Zenodo.
- Sklar, A. (1959): Fonctions de Répartition à n Dimensions et Leurs Marges. In *Publications de l'Institut Statistique de l'Université de Paris*, 8, pp. 229–231.
- Straja, S. R. (2004): Value at risk for long/short positions. Available online at <http://www.fintools.com/wp-content/uploads/2012/02/VaRLongShortPortfolio.pdf>, updated on 2/1/2004, checked on 2/5/2022.
- Taylor, S. J.; Xu, J. (1997): The Incremental Volatility Information in One Million Foreign Exchange Quotations. In *Journal of Empirical Finance* 4, pp. 317–340. DOI: 10.1016/S0927-5398(97)00010-8.
- Trikha, R. (Ed.) (N.d.): Emerging Languages Overshadowed by Incumbents Java, Python in Coding Interviews. HackerRank. Available online at <https://blog.hackerrank.com/emerging-languages-still-overshadowed-by-incumbents-java-python-in-coding-interviews/>, checked on 12/24/2021.
- Tse, Y. K.; Tsui, A. K. C. (2002): A Multivariate Generalized Autoregressive Conditional Heteroscedasticity Model With Time-Varying Correlations. In *Journal of Business & Economic Statistics* 20 (3), pp. 351–362. DOI: 10.1198/073500102288618496.
- User jcthomass (2022): Same input and output in a callback. Plotly Community Forum. Available online at <https://dash.plotly.com/datatable/conditional-formatting>, updated on 4/1/2019, checked on 1/19/2022.

User Jwolf (2022): Infinite loop (or Python app) plus Dash. Plotly Community Forum. Available online at <https://dash.plotly.com/datatable/conditional-formatting>, updated on 7/1/2021, checked on 1/19/2022.

User thedirtyfew (2021): `callback_serverside_output.py`. Available online at [https://github.com/thedirtyfew/dash-extensions/blob/master/examples/callback\\_serverside\\_output.py](https://github.com/thedirtyfew/dash-extensions/blob/master/examples/callback_serverside_output.py), updated on 11/5/2021, checked on 1/19/2022.

Voges, M.; Leschinski, C.; Sibbertsen, P. (2019): Seasonal long memory in intra-day volatility and trading volume of Dow Jones stocks. In J. Chevallier, S. Goutte, D. Guerreiro, S. Saglio, B. Sanhaji (Eds.): *International Financial Markets*. Abingdon, Oxon, New York, NY : Routledge, 2019. | Series: Routledge advances in applied financial econometrics, volume 1: Routledge, pp. 200–224.

Wagner, N. F.; Marsh, T. A. (2005): Surprise Volume and Heteroskedasticity in Equity Market Returns. In *SSRN Journal*. DOI: 10.2139/ssrn.591206.

Walther, U. (2020): Lecture 5: Estimating Market Risk. lecture notes. Berlin School of Economics and Law. Berlin.

Wang, J. X.; Yang, M. (2017): Conditional Volatility Persistence. In *SSRN Journal*. DOI: 10.2139/ssrn.3080693.

Wood, R. A.; McInish, T. H.; Ord, J. K. (1985): An investigation of transaction data for NYSE stocks. In *Journal of Finance* 40 (3), pp. 723–739. DOI: 10.1111/j.1540-6261.1985.tb04996.x.

Zoubir, L. (2017): A brief history of time series analysis. Stockholm University. Available online at <https://www.statistics.su.se/english/research/time-series-analysis/a-brief-history-of-time-series-analysis-1.259451>, updated on 10/31/2017, checked on 1/30/2022.

## Appendix A

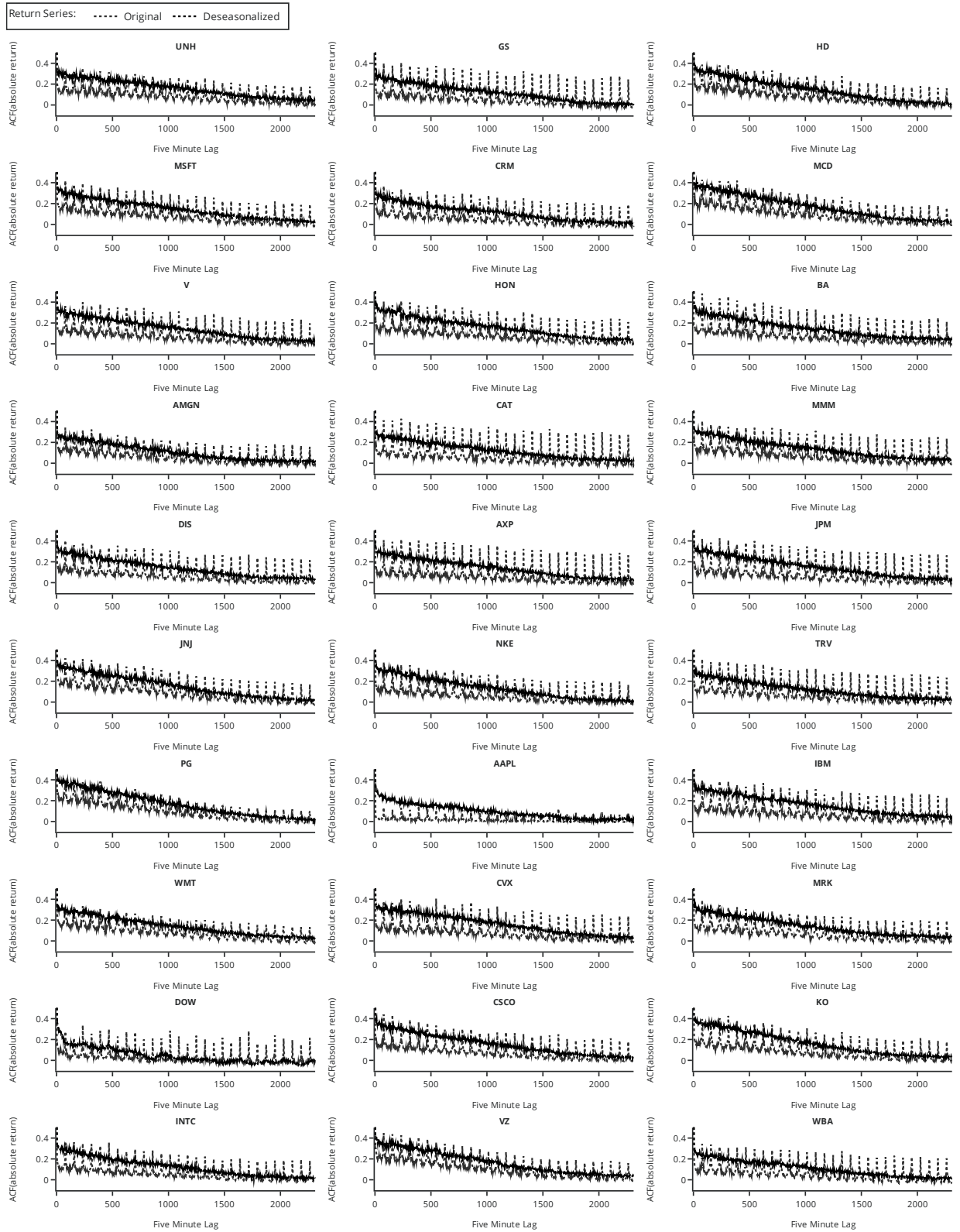
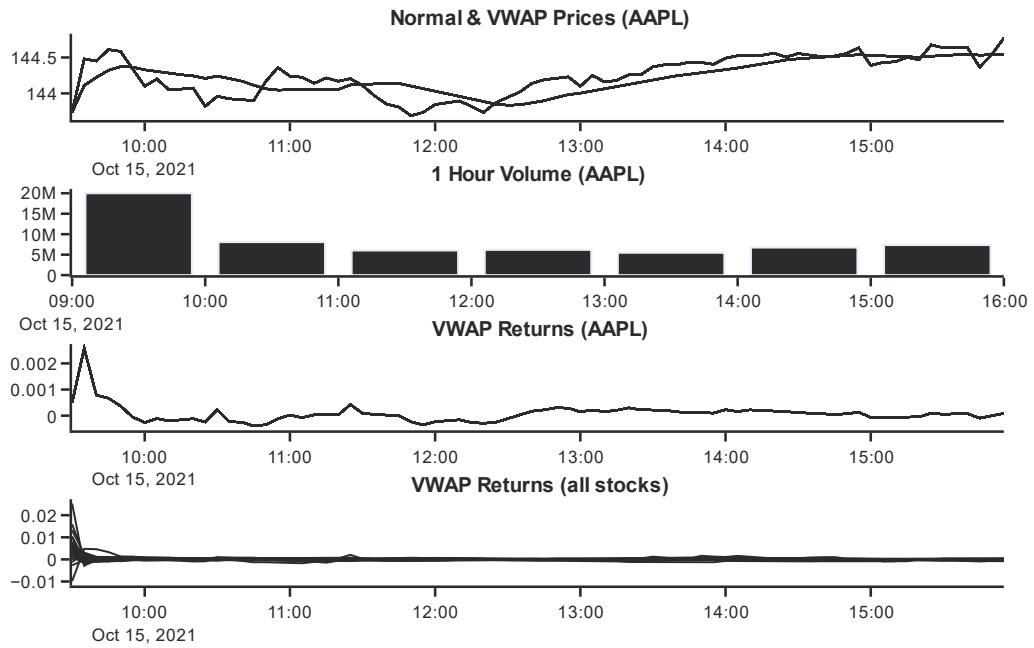


Figure 13: 30-day ACF plots of research sample.



**Figure 14:** Subplot 1 - raw prices and smoothed VWAP prices, Subplot 2 – 1h volume, Subplot 3 - example VWAP return series, Subplot 4 - VWAP return series of entire sample.

**Table 7:** Backtest Output for the calm scenario using t-distributed random MCM draws.

| Iteration | date             | P/L      | VaR loss | dcc alpha | dcc beta | Parameter Optimization | Model Input Days | Degrees of Freedom |
|-----------|------------------|----------|----------|-----------|----------|------------------------|------------------|--------------------|
| 0         | 08/10/2021 09:35 | -675.972 | -793.670 | 0.005484  | 0.340952 | 2.686                  | 5                | 56.255             |
| 1         | 08/10/2021 09:40 | -38.962  | -739.470 | 0.005767  | 0.353932 | 3.958                  | 5                | 66.193             |
| 2         | 08/10/2021 09:45 | -150.692 | -710.790 | 0.005592  | 0.353766 | 3.844                  | 5                | 51.544             |
| 3         | 08/10/2021 09:50 | -182.365 | -698.460 | 0.005844  | 0.318201 | 3.816                  | 5                | 59.990             |
| 4         | 08/10/2021 09:55 | -152.075 | -680.700 | 0.005786  | 0.330576 | 3.902                  | 5                | 57.573             |
| ...       | ...              |          |          |           |          |                        |                  |                    |
| 384       | 14/10/2021 15:35 | 89.8553  | -746.21  | 0.004935  | 0.200289 | 2.825                  | 5                | 25.97              |
| 385       | 14/10/2021 15:40 | 347.074  | -697.7   | 0.004959  | 0.202694 | 2.568                  | 5                | 42.115             |
| 386       | 14/10/2021 15:45 | 347.629  | -707.4   | 0.00517   | 0.200946 | 2.682                  | 5                | 30.366             |
| 387       | 14/10/2021 15:50 | 502.396  | -698.84  | 0.005072  | 0.205022 | 2.688                  | 5                | 29.649             |
| 388       | 14/10/2021 15:55 | 297.101  | -714.96  | 0.004852  | 0.2176   | 2.918                  | 5                | 29.947             |



**Figure 15:** VaR Exceedance occurrences across backtest window.



## Appendix B

### DCC Garch Tool

Instructions

Live Simulation

Adverse Scenario Analysis

#### Parameters

Bid-Ask Spread (%)

Maximum Ordersize

Orders Per Iteration

Margin Requirement On Short Positions (%)

Date

Resume

Stop

Reset

#### Simulation Status

Status Stopped

Computing Time per Iteration:

Param. Opt.: 1.37 seconds

Risk: 3.73 seconds

Total: 4.32 seconds

VaR & ES

DCC Model Parameters

Position Table

DCC Matrix

Toggle All

Incl. Fl. Profits

#### Value at Risk

434.21

#### Expected Shortfall

545.37

#### DCC Model Parameters

a: 0.00057

b: 0.20434

1-a-b: 0.7951

#### Univariate GARCH Parameters

| Ticker   | AMGN            | AXP             | BA              | CAT             | CSCO            | CVX            | DIS             | DOW             | HD              |
|----------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|-----------------|-----------------|
| omega    | 0.059 p:0.35386 | 0.634 p:0.0011  | 0.037 p:0.12035 | 0.405 p:0.16759 | 0.045 p:0.25432 | 0.01 p:0.27066 | 0.736 p:0.39671 | 1.871 p:0.70141 | 0.937 p:0.56744 |
| alpha[1] | 0.033 p:0.16961 | 0.036 p:0.28708 | 0.025 p:0.23408 | 0.082 p:0.06118 | 0.042 p:0.12937 | 0.0 p:1.0      | 0.029 p:0.64354 | 0.0 p:1.0       | 0.038 p:0.59752 |
| beta[1]  | 0.929 p:0.0     | 0.658 p:0.0     | 0.964 p:0.0     | 0.689 p:0.00014 | 0.902 p:0.0     | 0.996 p:0.0    | 0.473 p:0.44535 | 0.0 p:1.0       | 0.0 p:1.0       |
| nu       | 496.445 p:0.0   | 330.584 p:0.0   | 193.697 p:0.0   | 281.367 p:0.0   | 499.327 p:0.0   | 389.953 p:0.0  | 234.478 p:0.0   | 204.474 p:0.0   | 120.796 p:0.0   |

Figure 16: App Tab 1 – (1/3).

| Ticker | Price  | Size | Notional  | Trade Profit | Inv. Val. | P/L     | Div. Effect |
|--------|--------|------|-----------|--------------|-----------|---------|-------------|
| V      | 227.25 | -314 | -71356.5  | 35.72        | 75.46     | 111.18  | 0.4         |
| AXP    | 159.48 | -331 | -52787.88 | 26.27        | -238.32   | -212.05 | 0.31        |
| HD     | 329.58 | -132 | -43504.56 | 21.8         | 97.68     | 119.48  | 0.25        |
| MMM    | 190.32 | -89  | -16938.48 | 8.48         | 28.48     | 36.96   | 0.01        |
| WBA    | 51.16  | -144 | -7366.32  | 3.68         | -0.72     | 2.96    | 0.07        |
| DIS    | 181.54 | -33  | -5990.98  | 8.44         | 0.15      | 8.59    | -0.01       |
| CSCO   | 59.09  | 41   | 2422.69   | 1.87         | 0.31      | 2.18    | 0           |
| WMT    | 147.88 | 24   | 3549.24   | 1.79         | -24.6     | -22.81  | -0.01       |
| HON    | 225.12 | 25   | 5628      | 34.89        | 46.36     | 81.25   | -0.13       |
| JNJ    | 172.86 | 65   | 11235.58  | 41.18        | -253.8    | -212.62 | -0.12       |
| JPM    | 161.08 | 74   | 11920.29  | 46.42        | -77.22    | -30.8   | -0.12       |
| DOW    | 62.08  | 276  | 17134.08  | 8.53         | 77.28     | 85.81   | 0.01        |
| VZ     | 55.24  | 326  | 18009.87  | 9.01         | -8.15     | 0.86    | 0.11        |
| CAT    | 209    | 97   | 20273     | 10.13        | 14.55     | 24.68   | 0.07        |
| CVX    | 97.82  | 236  | 23085.52  | 14.87        | -5.78     | 9.09    | 0.05        |
| PG     | 143.69 | 212  | 30462.28  | 15.29        | -111.3    | -96.01  | -0.02       |
| BA     | 214.66 | 195  | 41858.7   | 20.93        | 0         | 20.93   | 0.23        |
| UNH    | 419.35 | 128  | 53676.8   | 26.87        | -73.08    | -46.21  | 0.11        |
| AMGN   | 221.36 | 310  | 68621.6   | 34.32        | -26.1     | 8.22    | 0.18        |
| TRV    | 156.89 | 446  | 69972.94  | 35.09        | -192.9    | -157.81 | 0.15        |

Figure 17: App Tab 1 - (2/3).

## Dynamic Conditional Correlation Matrix

|      | AMGN  | AXP   | BA    | CAT   | CSCO | CVX   | DIS  | DOW   | HD   | HON   | JNJ   | JPM   | MMM   | PG    | TRV   | UNH  | V     | VZ   | WBA  | WMT |
|------|-------|-------|-------|-------|------|-------|------|-------|------|-------|-------|-------|-------|-------|-------|------|-------|------|------|-----|
| AMGN | 1     |       |       |       |      |       |      |       |      |       |       |       |       |       |       |      |       |      |      |     |
| AXP  | -0.09 | 1     |       |       |      |       |      |       |      |       |       |       |       |       |       |      |       |      |      |     |
| BA   | -0.03 | 0.41  | 1     |       |      |       |      |       |      |       |       |       |       |       |       |      |       |      |      |     |
| CAT  | -0.08 | 0.24  | 0.37  | 1     |      |       |      |       |      |       |       |       |       |       |       |      |       |      |      |     |
| CSCO | 0.11  | 0     | 0.06  | 0.12  | 1    |       |      |       |      |       |       |       |       |       |       |      |       |      |      |     |
| CVX  | -0.1  | 0.31  | 0.38  | 0.35  | 0.08 | 1     |      |       |      |       |       |       |       |       |       |      |       |      |      |     |
| DIS  | 0.07  | 0.19  | 0.25  | 0.18  | 0.06 | 0.26  | 1    |       |      |       |       |       |       |       |       |      |       |      |      |     |
| DOW  | -0.05 | 0.33  | 0.3   | 0.37  | 0.11 | 0.42  | 0.2  | 1     |      |       |       |       |       |       |       |      |       |      |      |     |
| HD   | 0.1   | -0.04 | 0.03  | 0.16  | 0.12 | -0.04 | 0.13 | 0.04  | 1    |       |       |       |       |       |       |      |       |      |      |     |
| HON  | 0.06  | 0.25  | 0.34  | 0.45  | 0.12 | 0.22  | 0.15 | 0.37  | 0.14 | 1     |       |       |       |       |       |      |       |      |      |     |
| JNJ  | 0.37  | -0.1  | -0.07 | 0     | 0.18 | -0.09 | 0.03 | -0.03 | 0.18 | 0.04  | 1     |       |       |       |       |      |       |      |      |     |
| JPM  | -0.02 | 0.52  | 0.47  | 0.43  | 0.09 | 0.4   | 0.25 | 0.41  | 0    | 0.34  | -0.06 | 1     |       |       |       |      |       |      |      |     |
| MMM  | 0.1   | 0.11  | 0.17  | 0.36  | 0.2  | 0.12  | 0.08 | 0.29  | 0.18 | 0.44  | 0.14  | 0.21  | 1     |       |       |      |       |      |      |     |
| PG   | 0.15  | -0.18 | -0.15 | -0.11 | 0.15 | -0.1  | 0.04 | -0.11 | 0.13 | 0.02  | 0.39  | -0.14 | 0.15  | 1     |       |      |       |      |      |     |
| TRV  | -0.03 | 0.28  | 0.22  | 0.31  | 0.06 | 0.12  | 0.08 | 0.27  | 0.17 | 0.3   | -0.01 | 0.44  | 0.24  | -0.02 | 1     |      |       |      |      |     |
| UNH  | 0.14  | 0.08  | -0.06 | 0.06  | 0.11 | -0.02 | 0.05 | 0.07  | 0.11 | 0.03  | 0.27  | 0.07  | 0.15  | 0.12  | 0.08  | 1    |       |      |      |     |
| V    | -0.02 | 0.23  | 0.04  | -0.04 | 0.1  | 0.02  | 0.05 | 0.03  | 0.01 | -0.01 | 0.04  | 0.12  | -0.02 | 0.04  | -0.01 | 0.06 | 1     |      |      |     |
| VZ   | 0.15  | 0.07  | 0.08  | 0.15  | 0.24 | 0.12  | 0.15 | 0.1   | 0.06 | 0.14  | 0.25  | 0.17  | 0.22  | 0.32  | 0.16  | 0.2  | 0.15  | 1    |      |     |
| WBA  | 0.12  | 0.17  | 0.23  | 0.24  | 0.15 | 0.18  | 0.25 | 0.19  | 0.07 | 0.22  | 0.13  | 0.26  | 0.14  | 0.06  | 0.17  | 0.11 | -0.04 | 0.22 | 1    |     |
| WMT  | 0.24  | -0.1  | -0.11 | -0.03 | 0.15 | -0.05 | 0.05 | 0.02  | 0.3  | 0.01  | 0.25  | -0.12 | 0.11  | 0.31  | 0.04  | 0.19 | -0.03 | 0.16 | 0.04 | 1   |

## Historic Dynamic Conditional Correlation



Figure 18: App Tab 1 - (3/3).

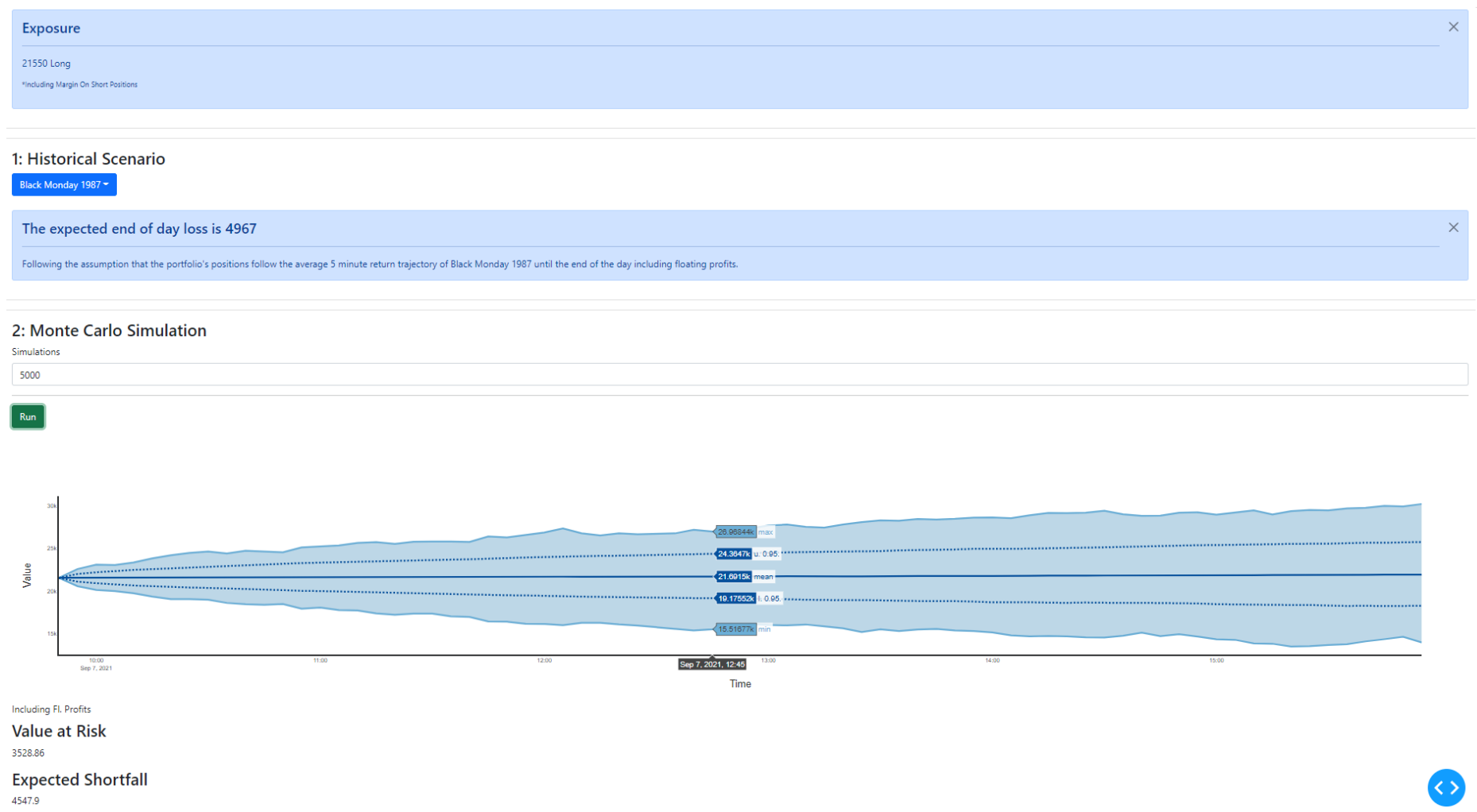


Figure 19: App Tab 2.

Legend:

- Inputs
- States
- Outputs
- Callbacks
- Dash Components
- Dash Component Attributes

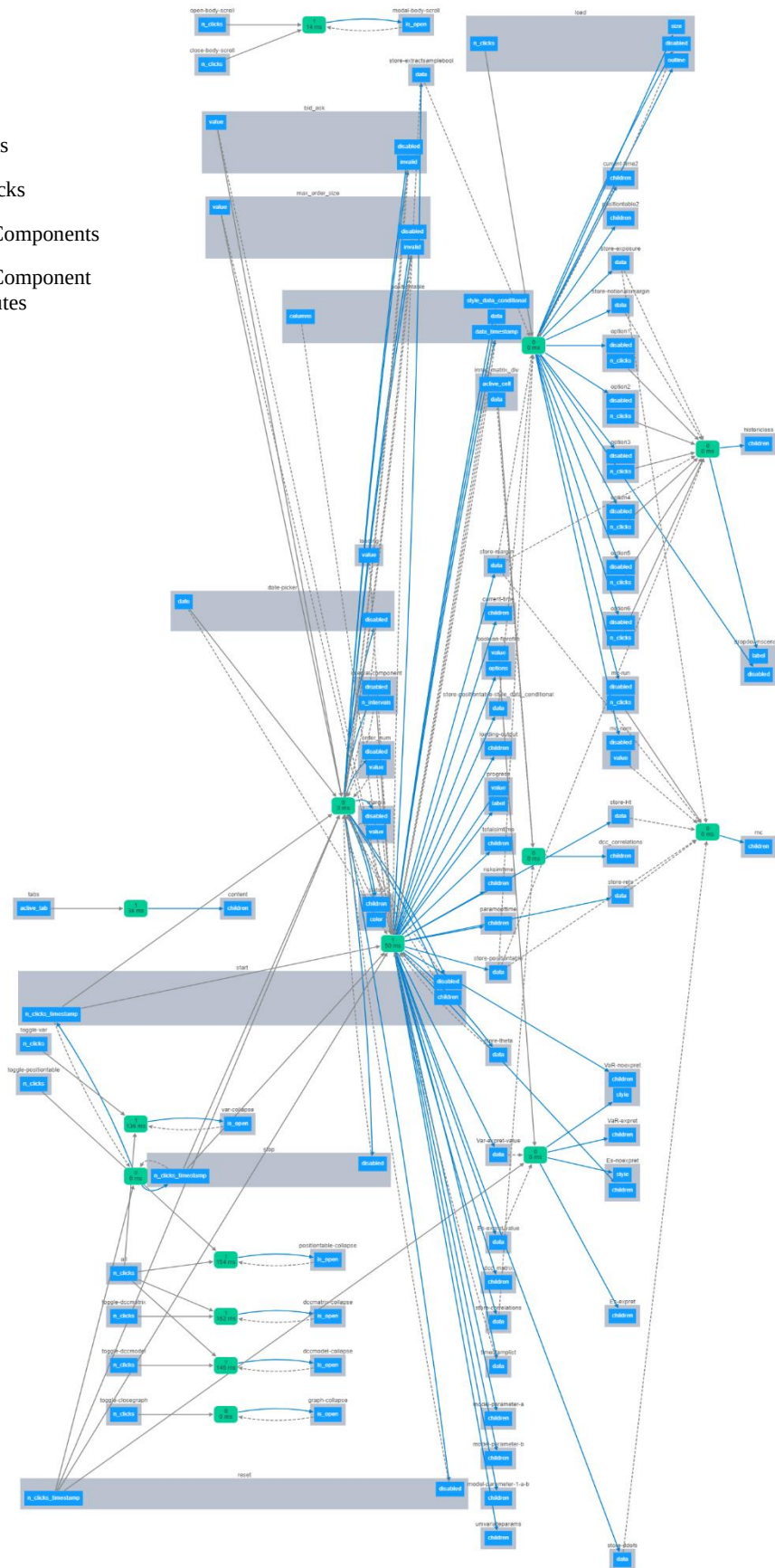


Figure 20: Flowchart of App from Dash debugger - process from left to right.

### Sworn declaration

I hereby formally declare that I have written the submitted Master's thesis entirely by myself without anyone else's assistance. Where I have drawn on literature or other sources, either in direct quotes, or in paraphrasing such material, I have referenced the original author or authors and the source in which it appeared.

I am aware that the use of quotations, or of close paraphrasing, from books, magazines, newspapers, the internet or other sources, which are not marked as such, will be considered as an attempt at deception, and that the thesis will be graded as a fail. In the event that I have submitted the dissertation - either in whole or in part - for examination within the framework of another examination, I have informed the examiners and the board of examiners of this fact.

12.02.2022

Berlin, Date



Signature